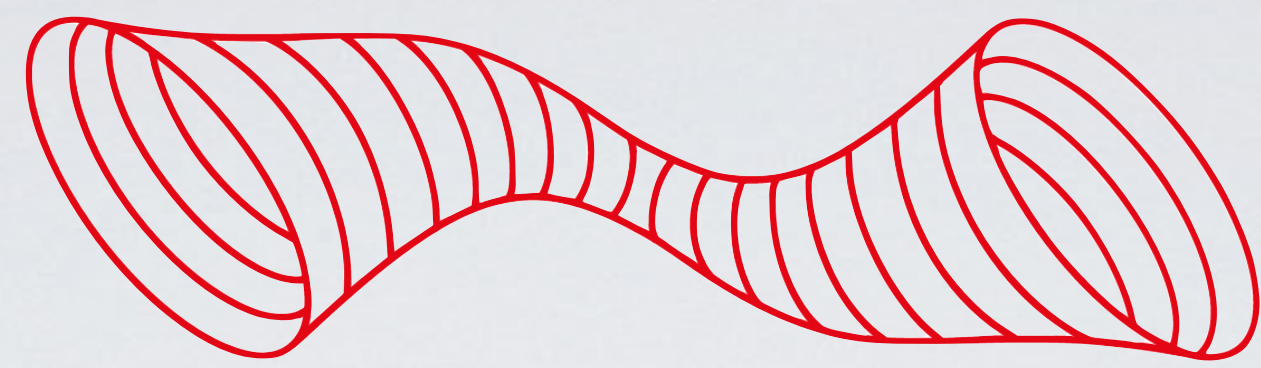
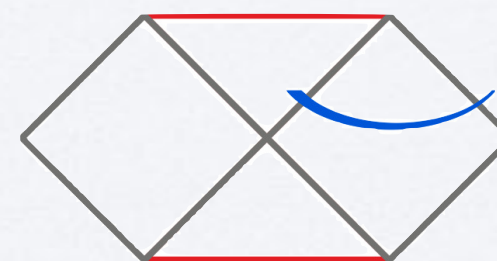




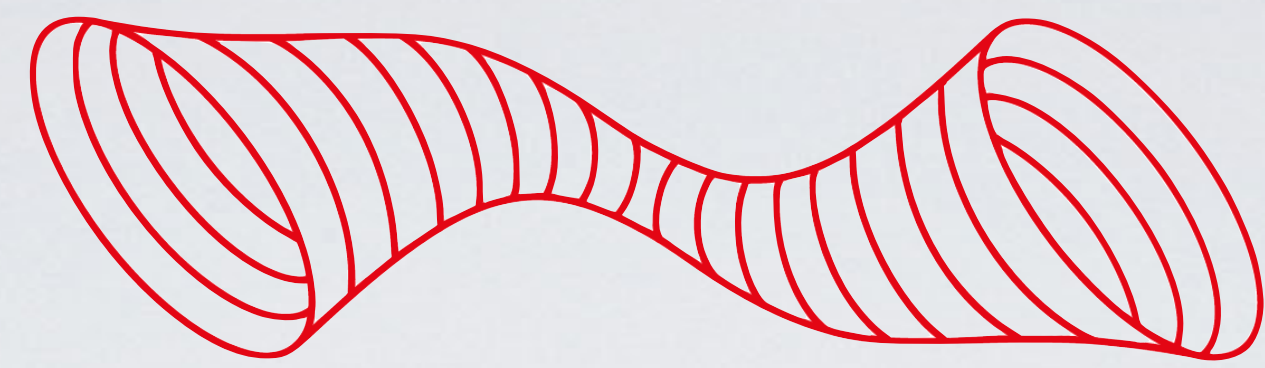
THE CENTER OF GRAVITY

FALLING INTO A BLACK HOLE

Rodrigo Panosso Macedo



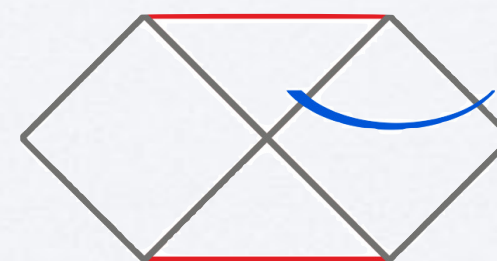
<https://hyperboloid.al>



THE CENTER OF GRAVITY

FALLING INTO A BLACK HOLE

Rodrigo Panosso Macedo



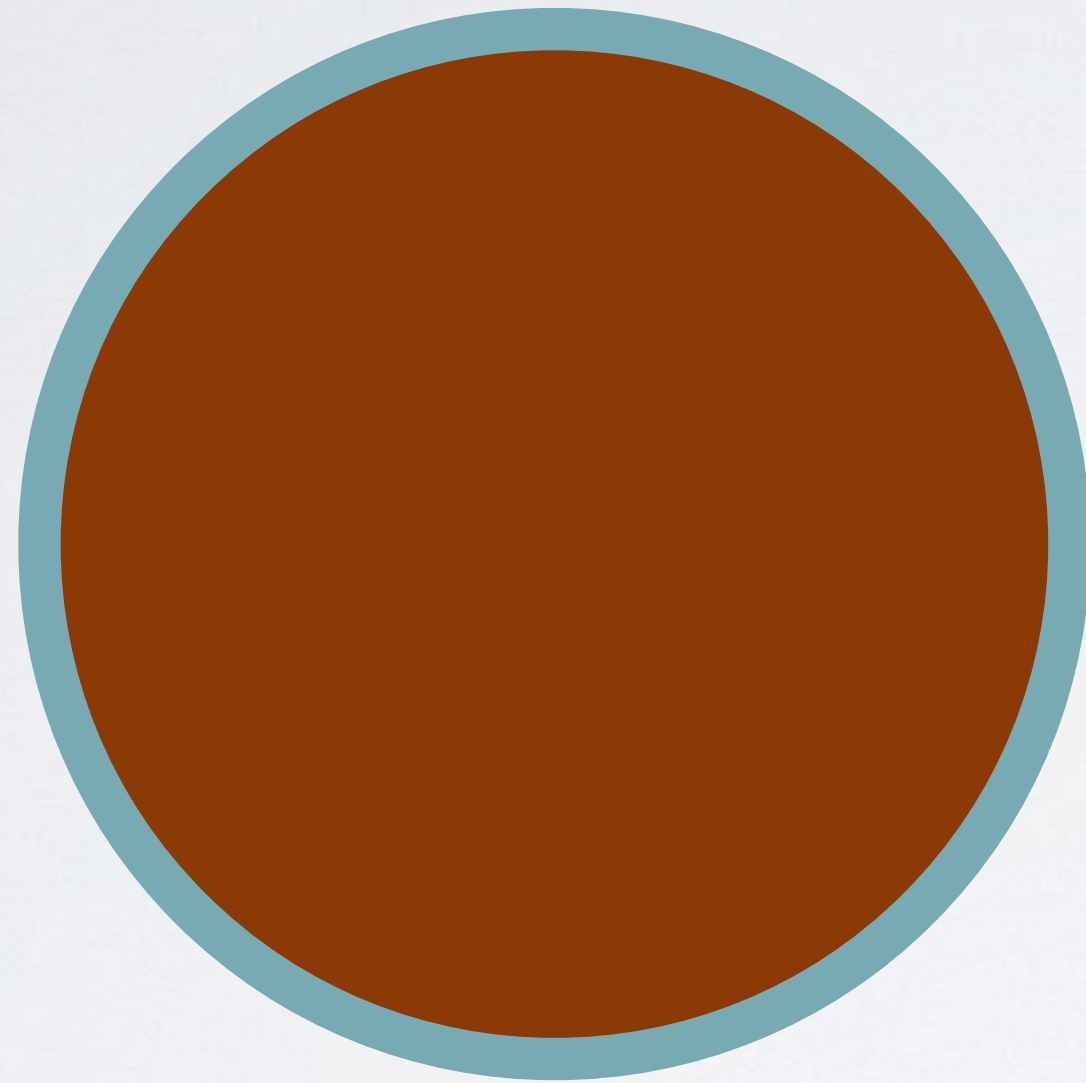
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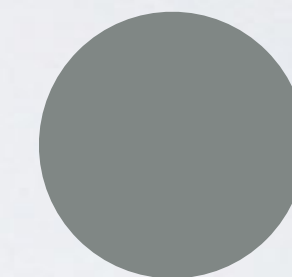
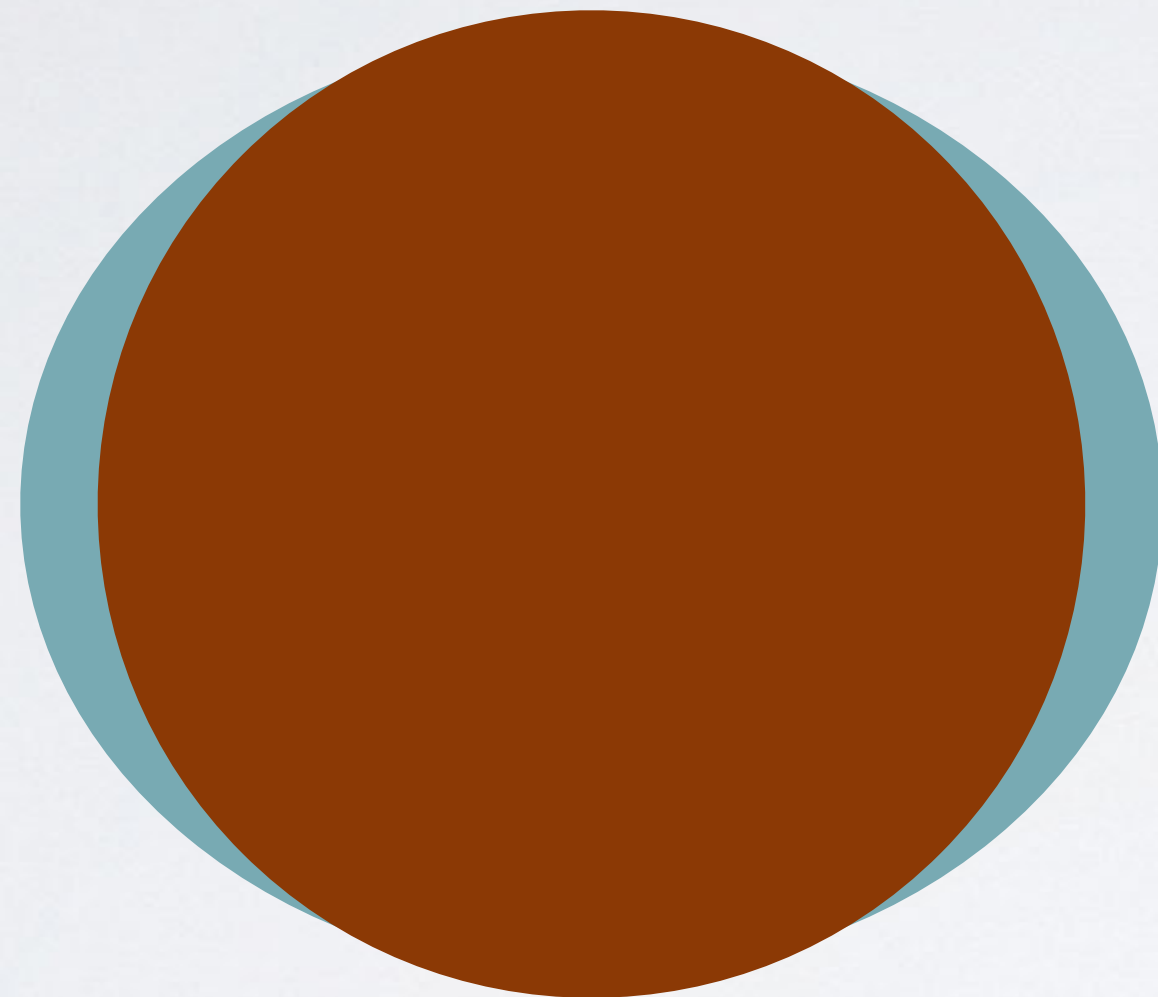




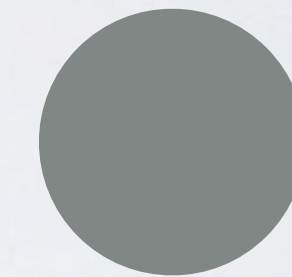
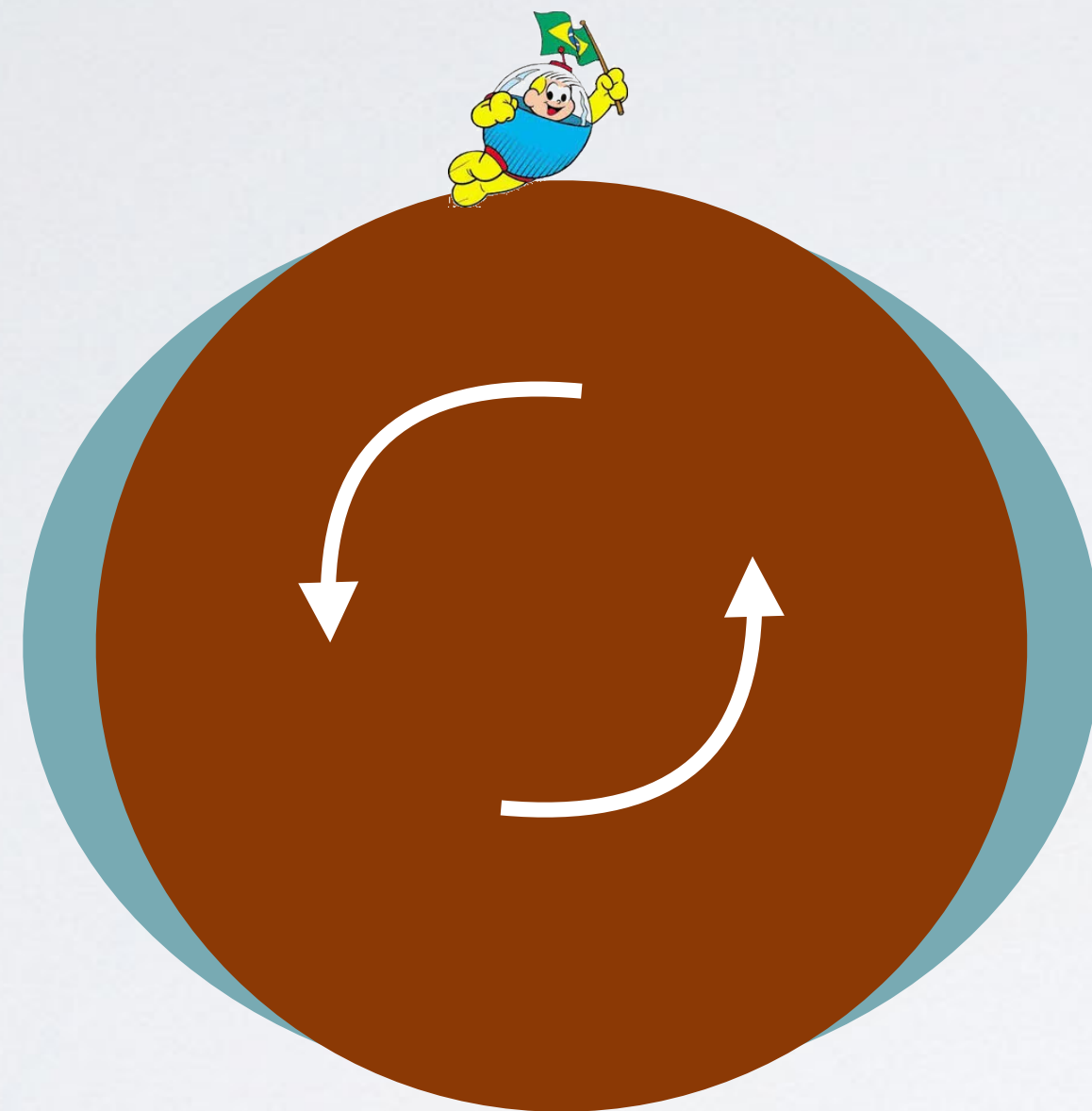
TIDAL EFFECT



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TIDAL EFFECT



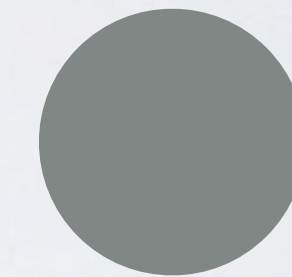
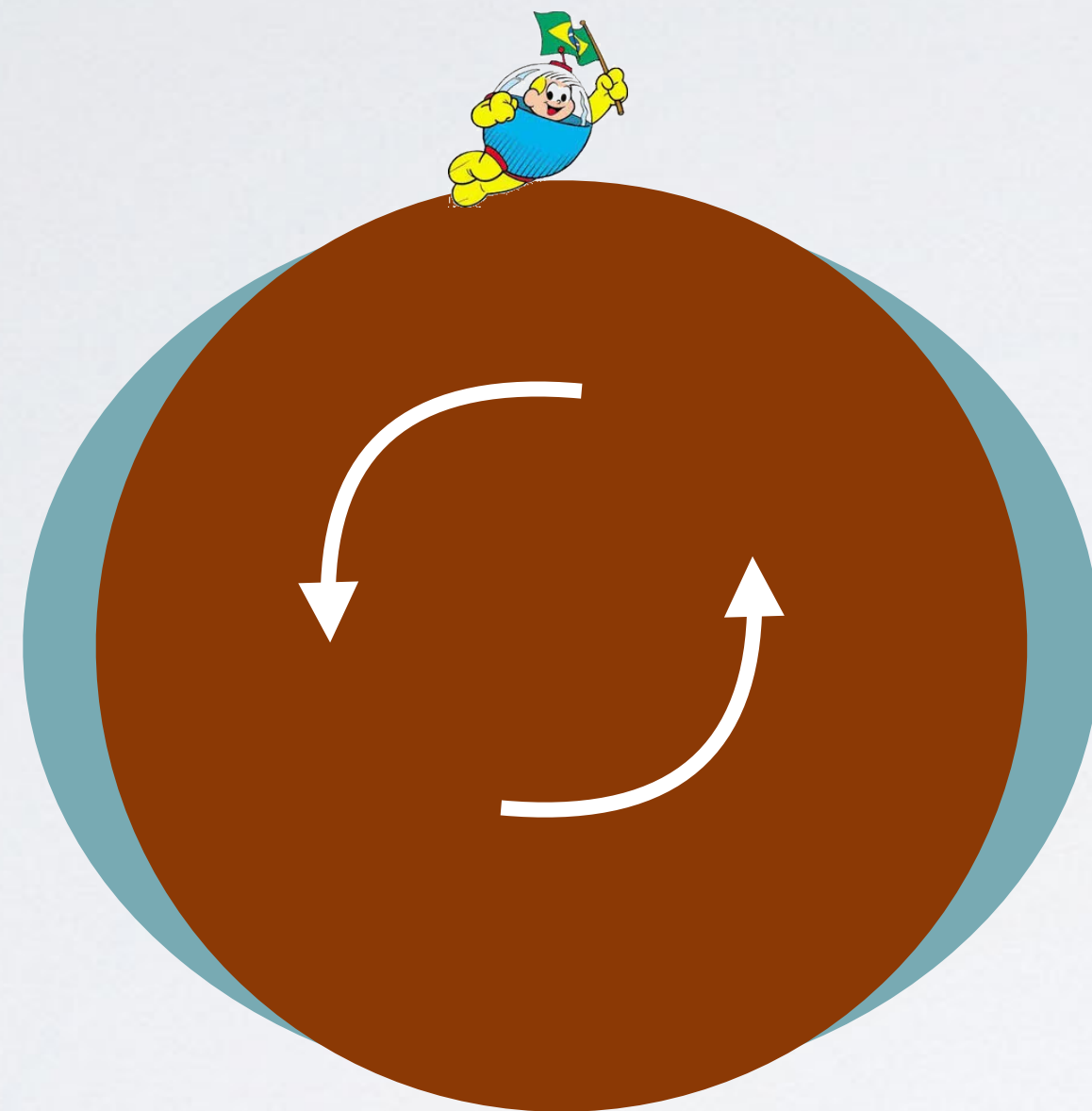
TIDAL EFFECT



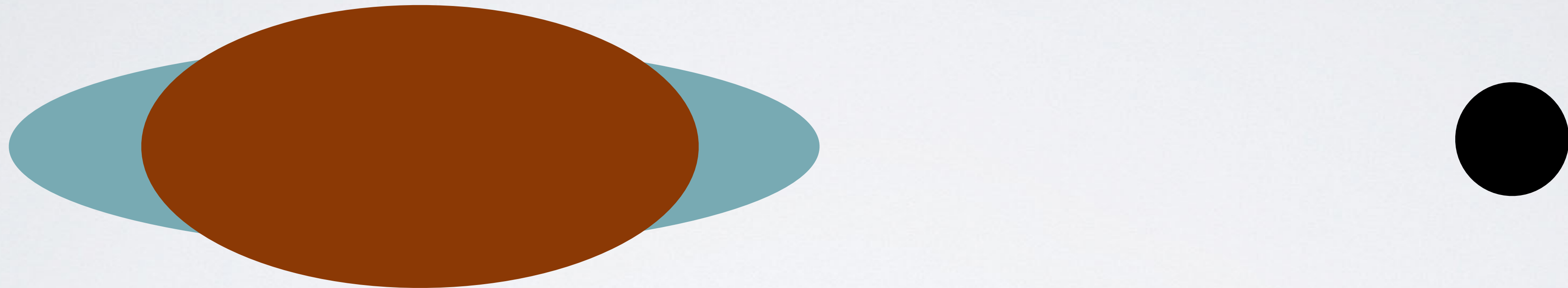
TIDAL EFFECT



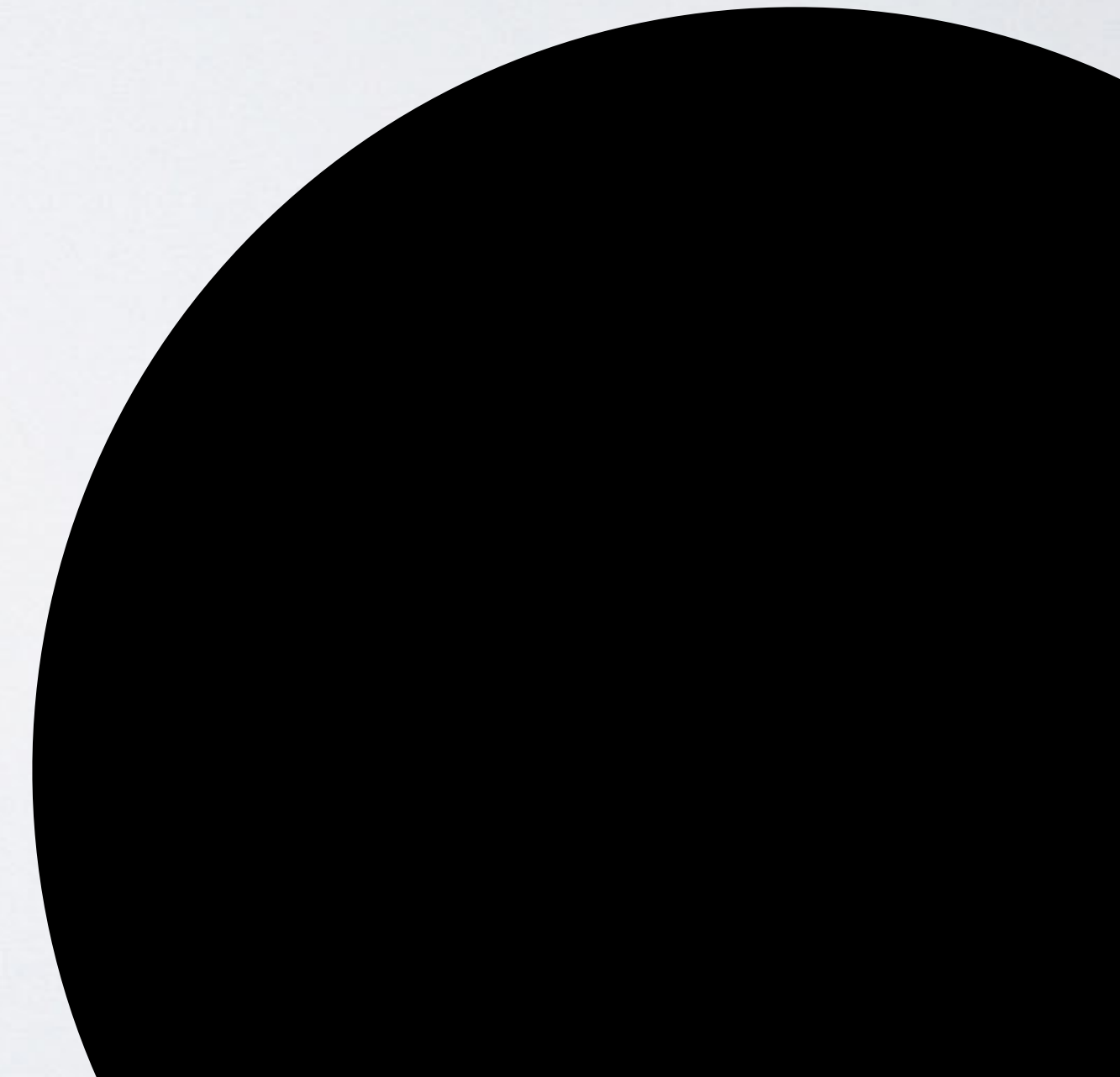
TIDAL EFFECT



TIDAL EFFECT FROM A BLACK HOLE



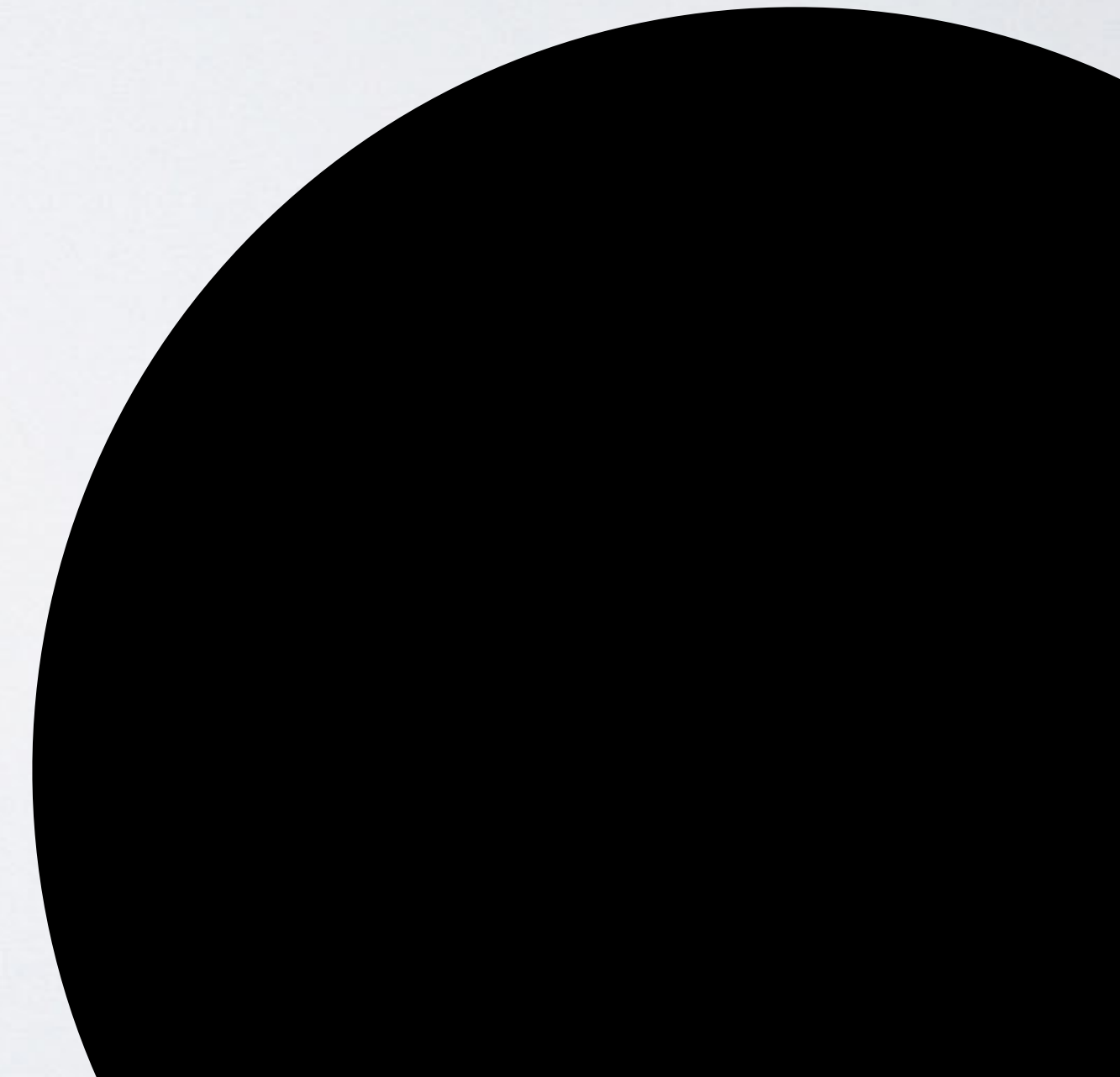
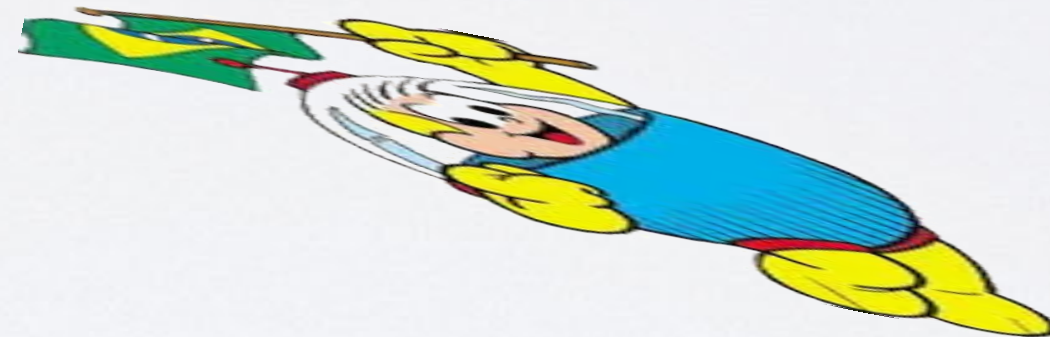
TIDAL EFFECT FROM A BLACK HOLE



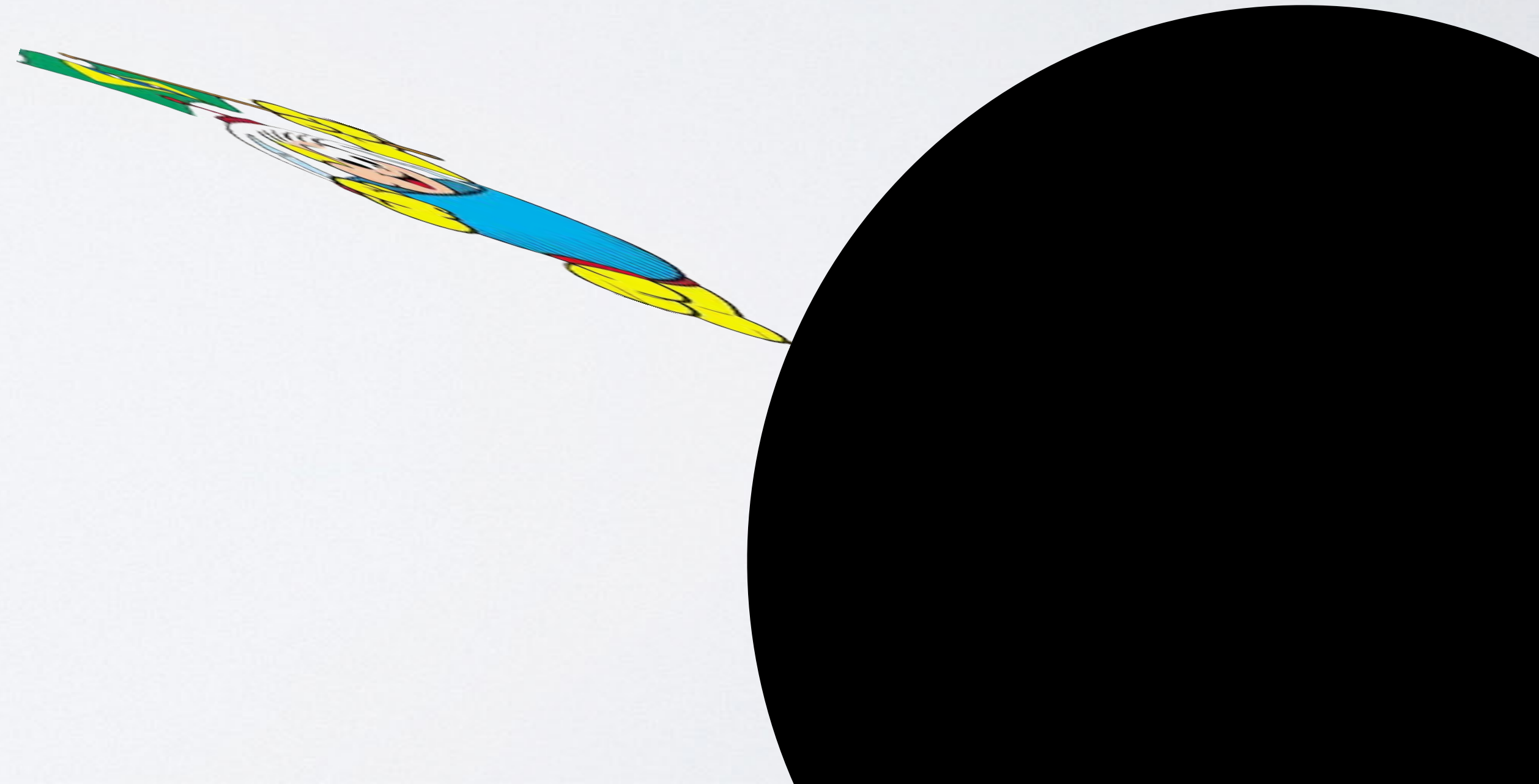
TIDAL EFFECT FROM A BLACK HOLE



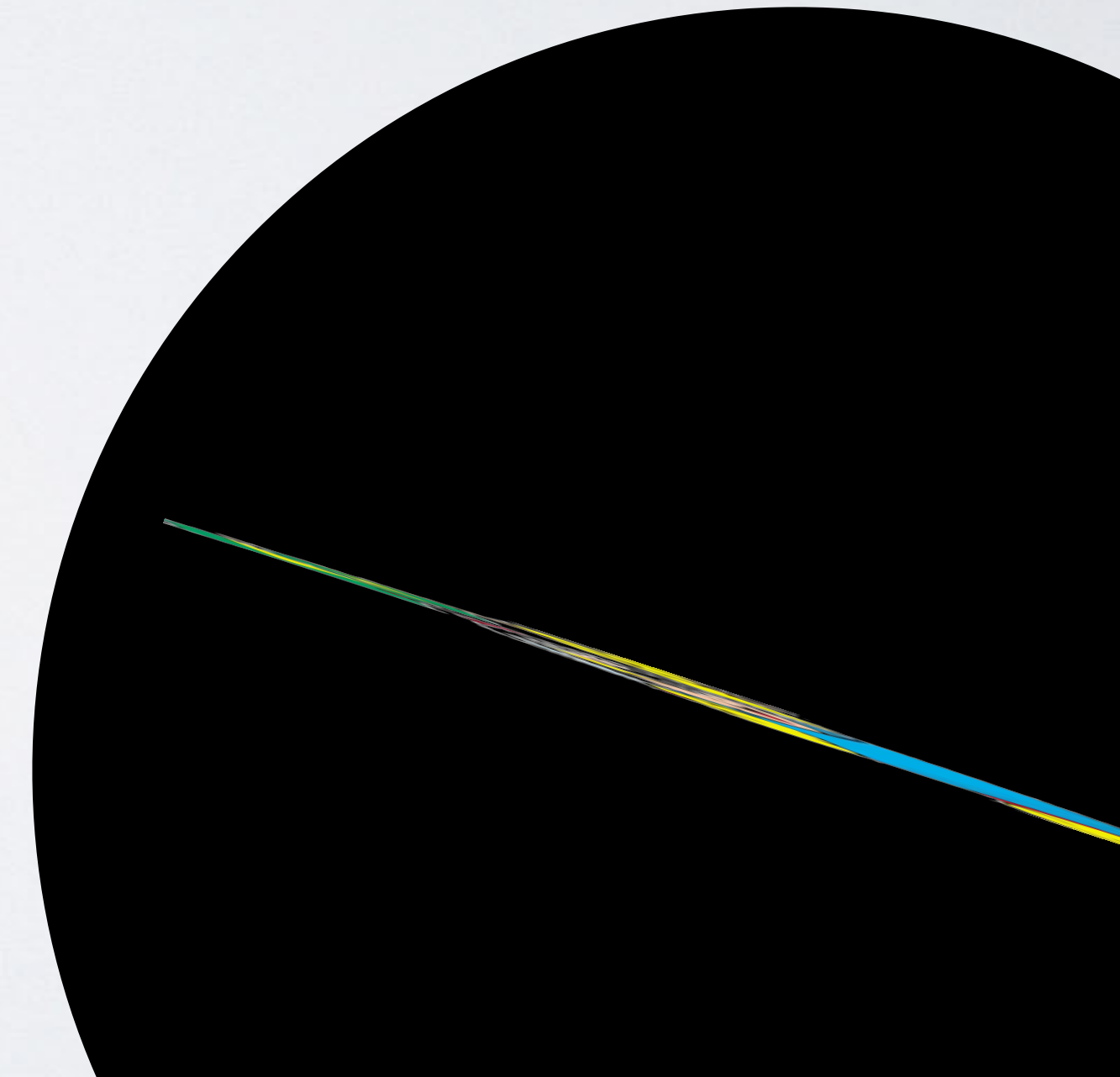
TIDAL EFFECT FROM A BLACK HOLE



TIDAL EFFECT FROM A BLACK HOLE

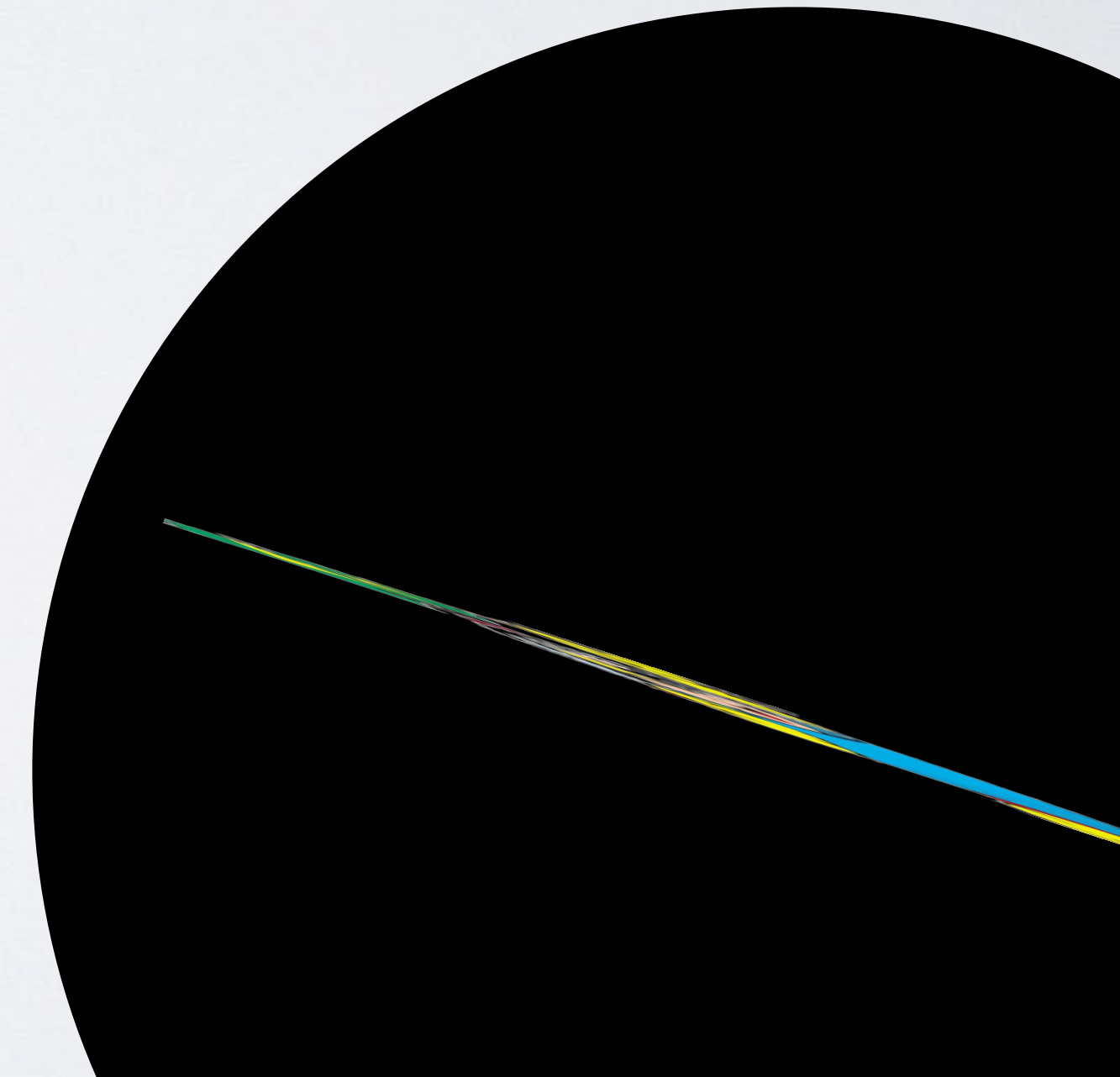


TIDAL EFFECT FROM A BLACK HOLE



TIDAL EFFECT FROM A BLACK HOLE

SPAGHETIFICATION



TIDAL EFFECT FROM A BLACK HOLE



TIDAL EFFECT FROM A BLACK HOLE



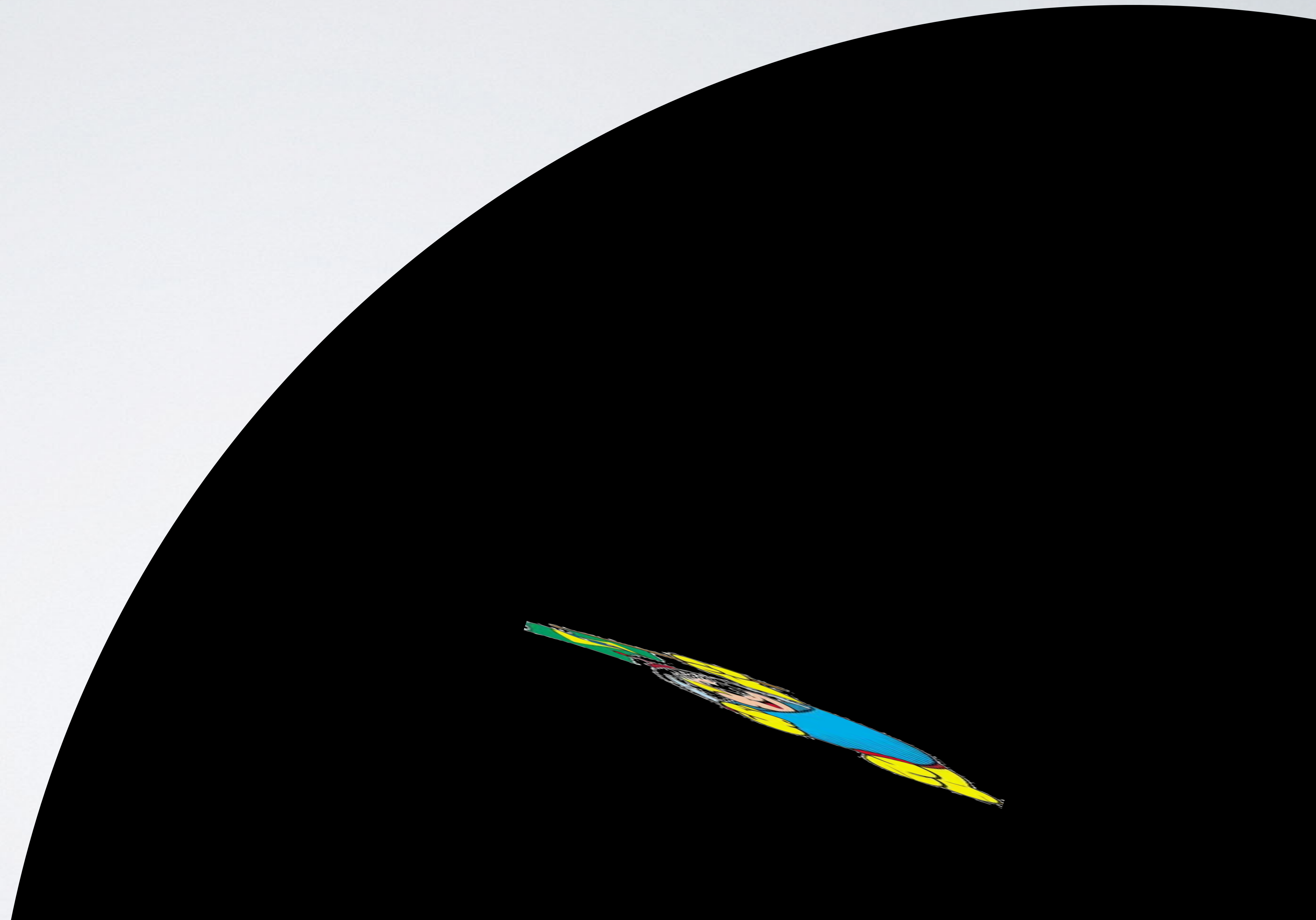
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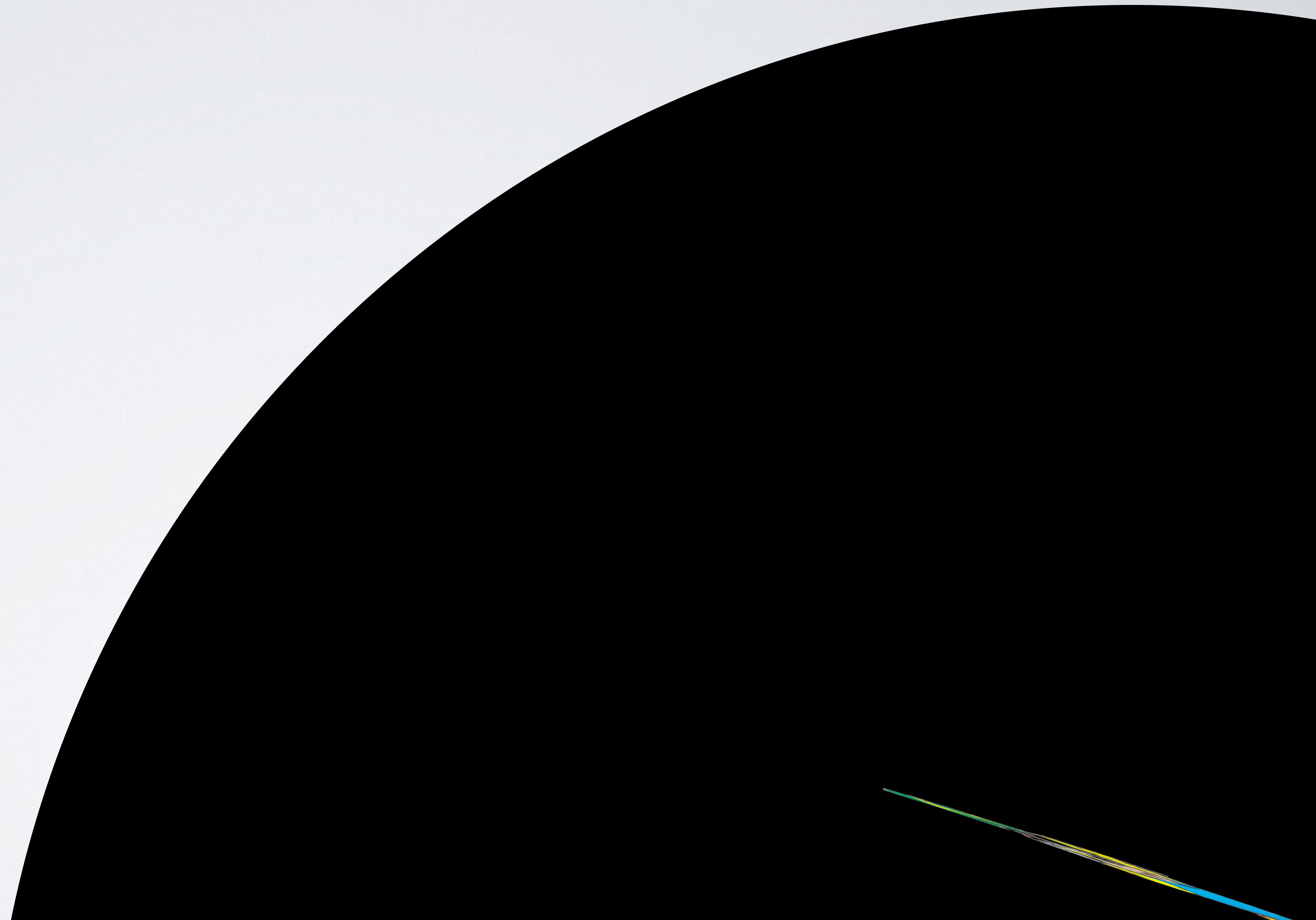
TIDAL EFFECT FROM A BLACK HOLE



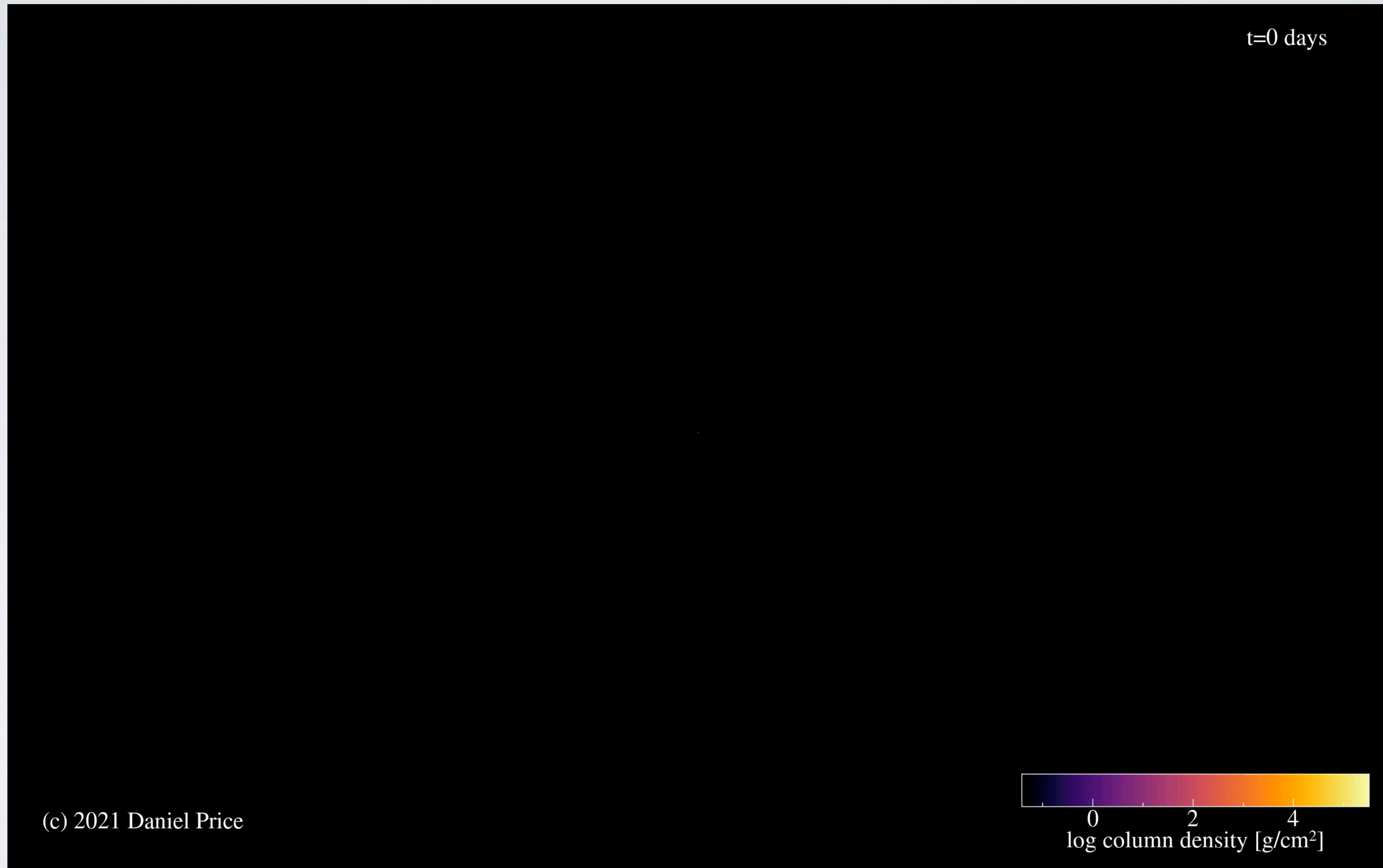
TIDAL EFFECT FROM A BLACK HOLE



TIDAL EFFECT FROM A BLACK HOLE

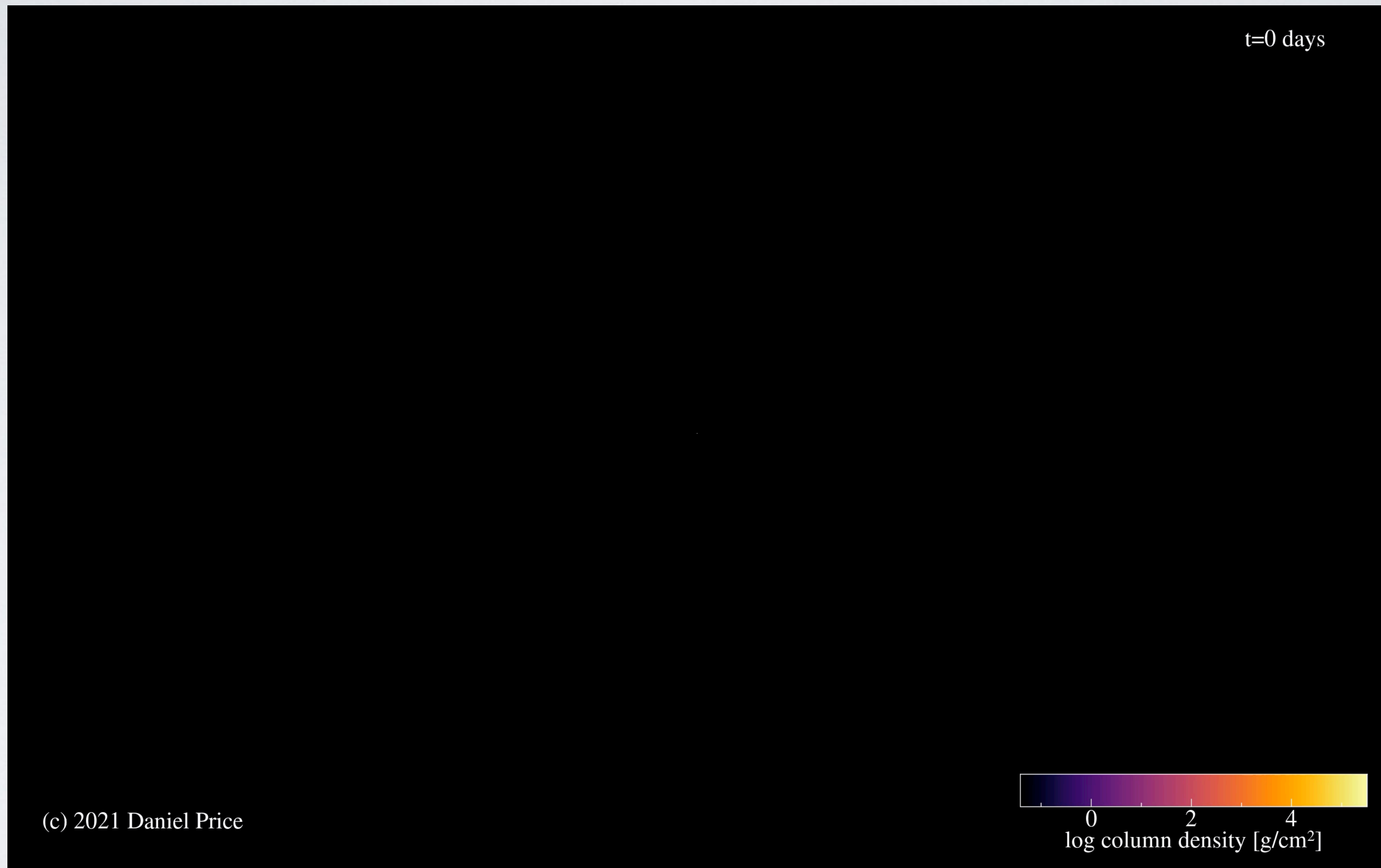


TIDAL DISRUPTION EVENTS



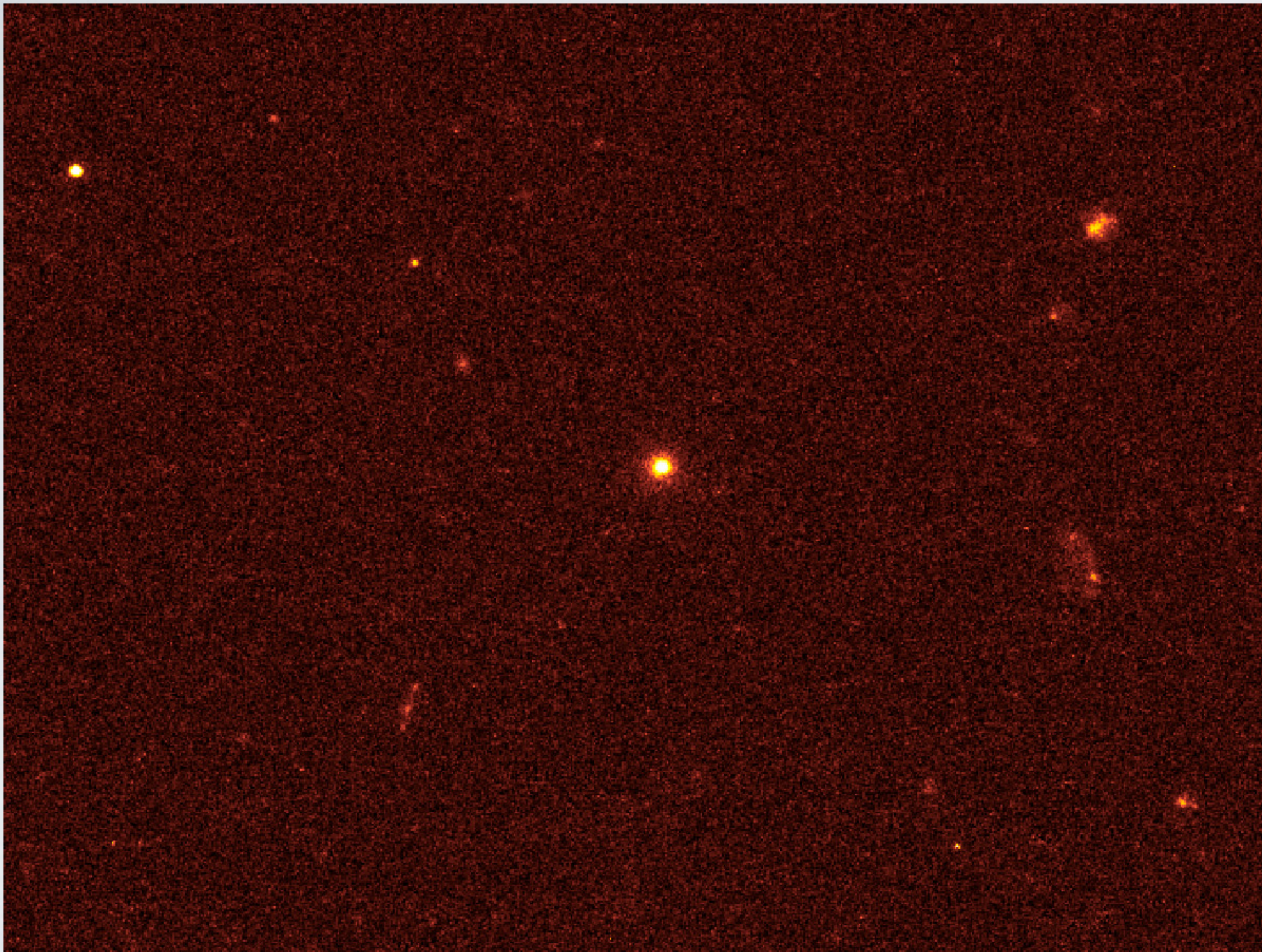
Price et al, ApJL 971 L46 (2024)

TIDAL DISRUPTION EVENTS



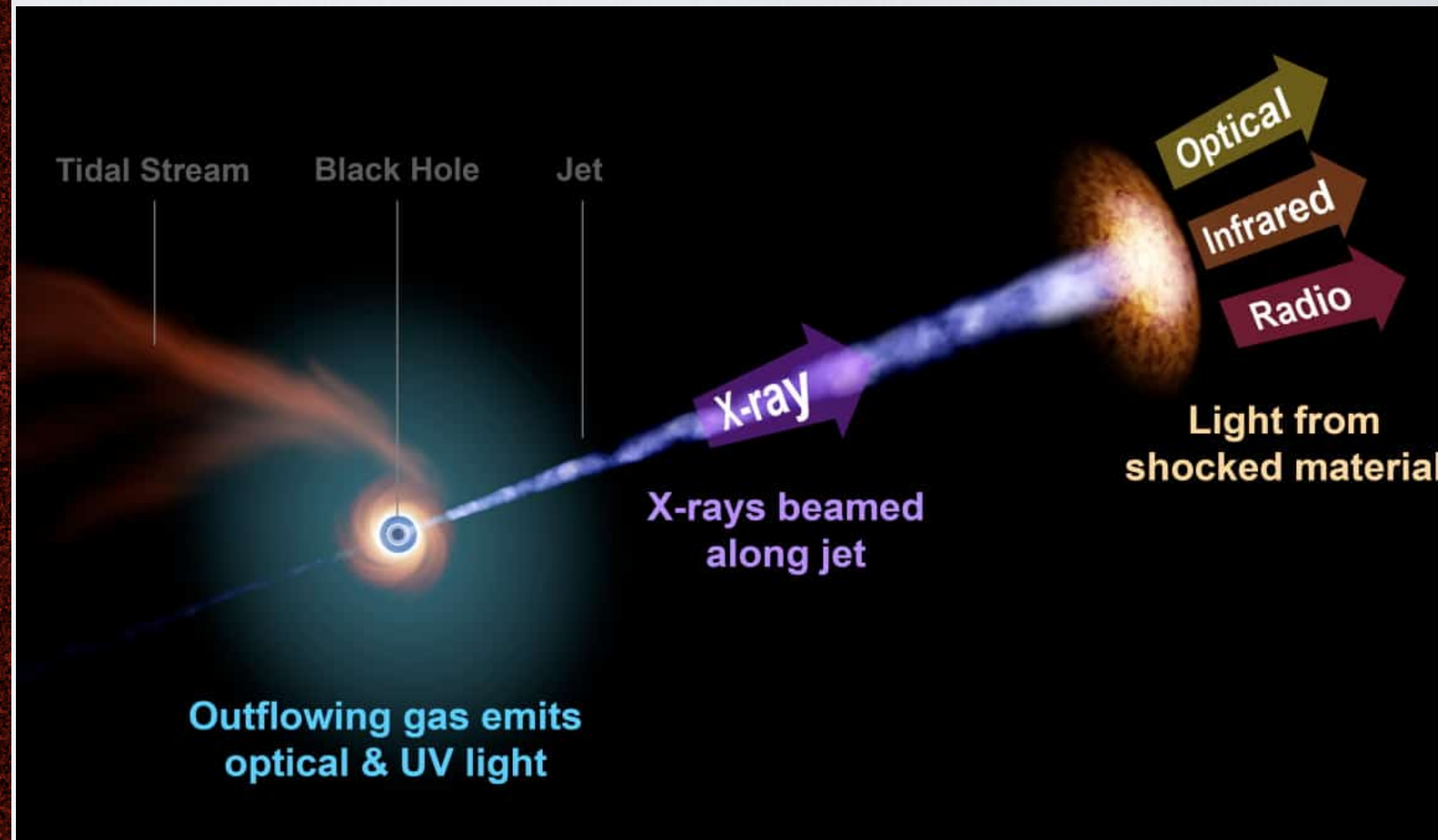
Price et al, ApJL 971 L46 (2024)

TIDAL DISRUPTION EVENTS



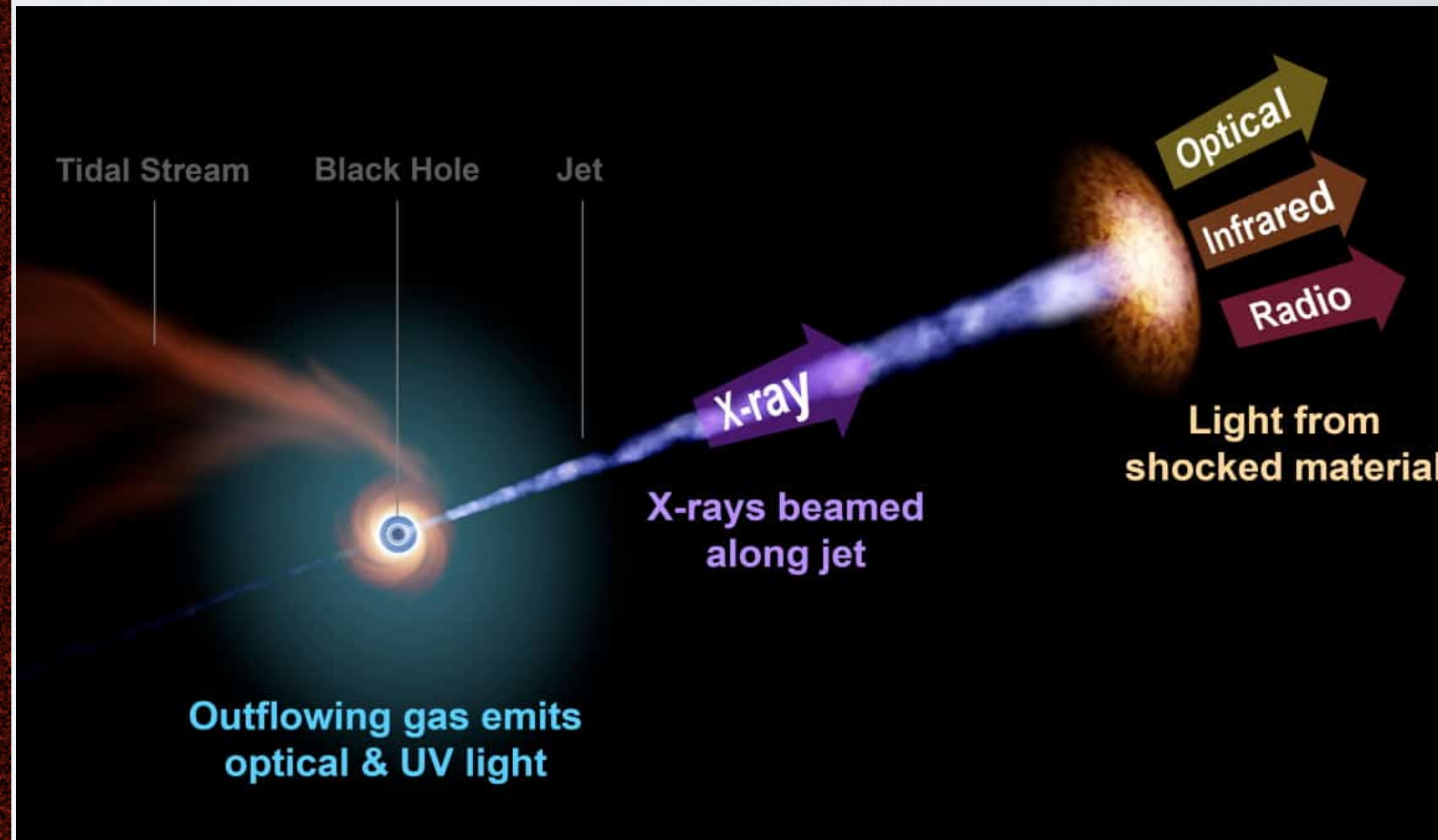
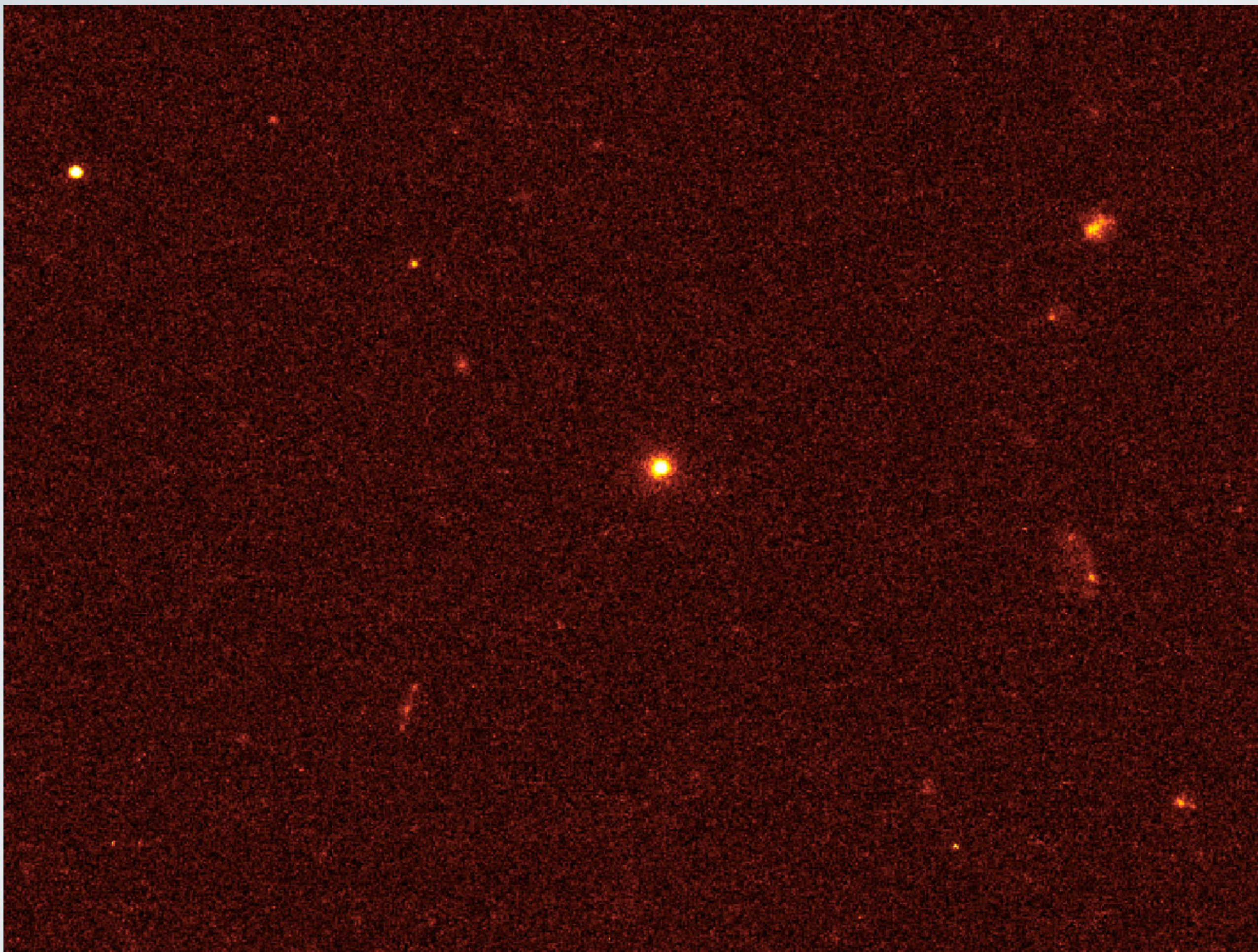
Swift J1644+57

By NASA, ESA, and A. Fruchter (STScI)



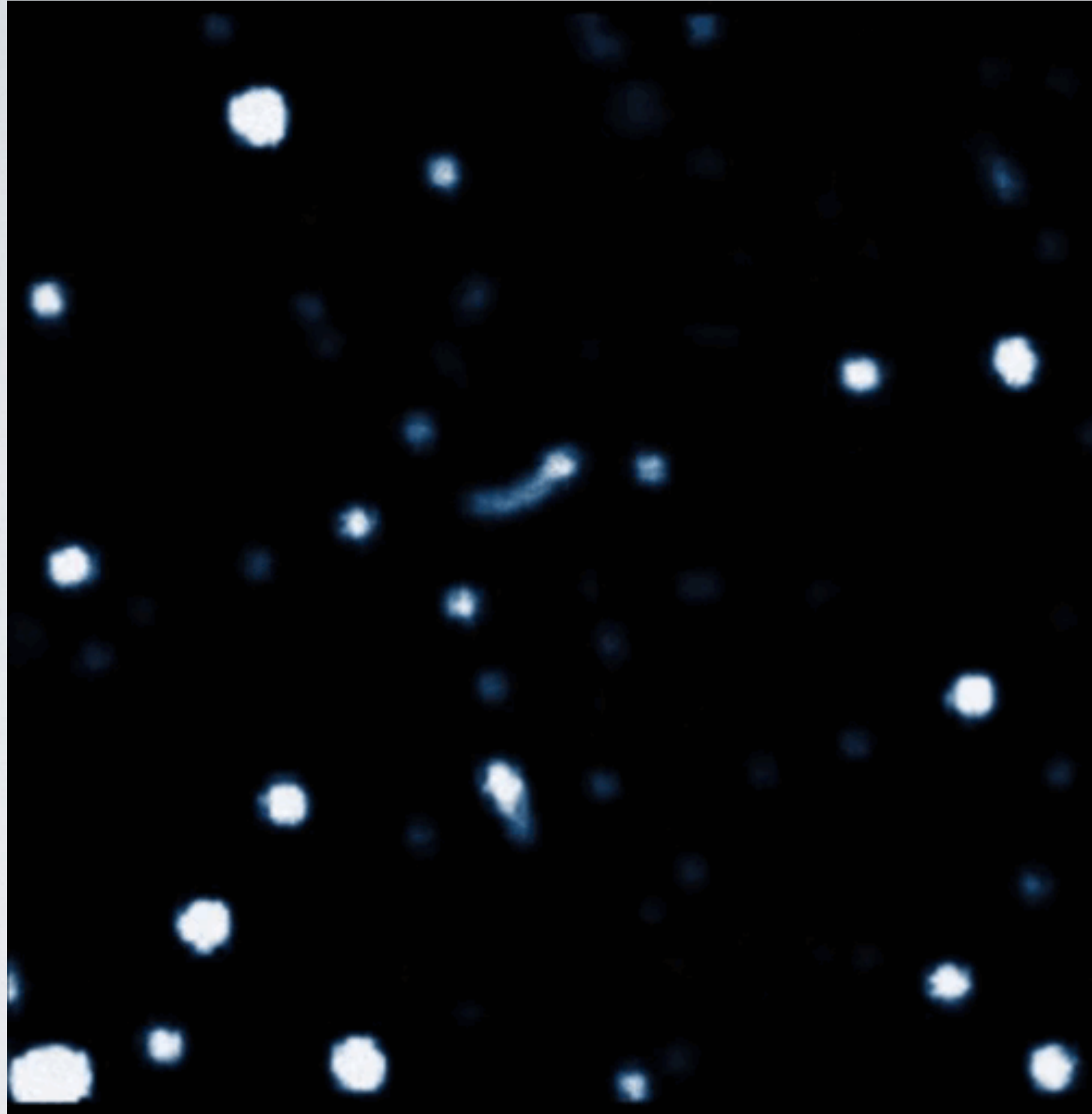
Zwicky Transient Facility/R Hurt
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TIDAL DISRUPTION EVENTS

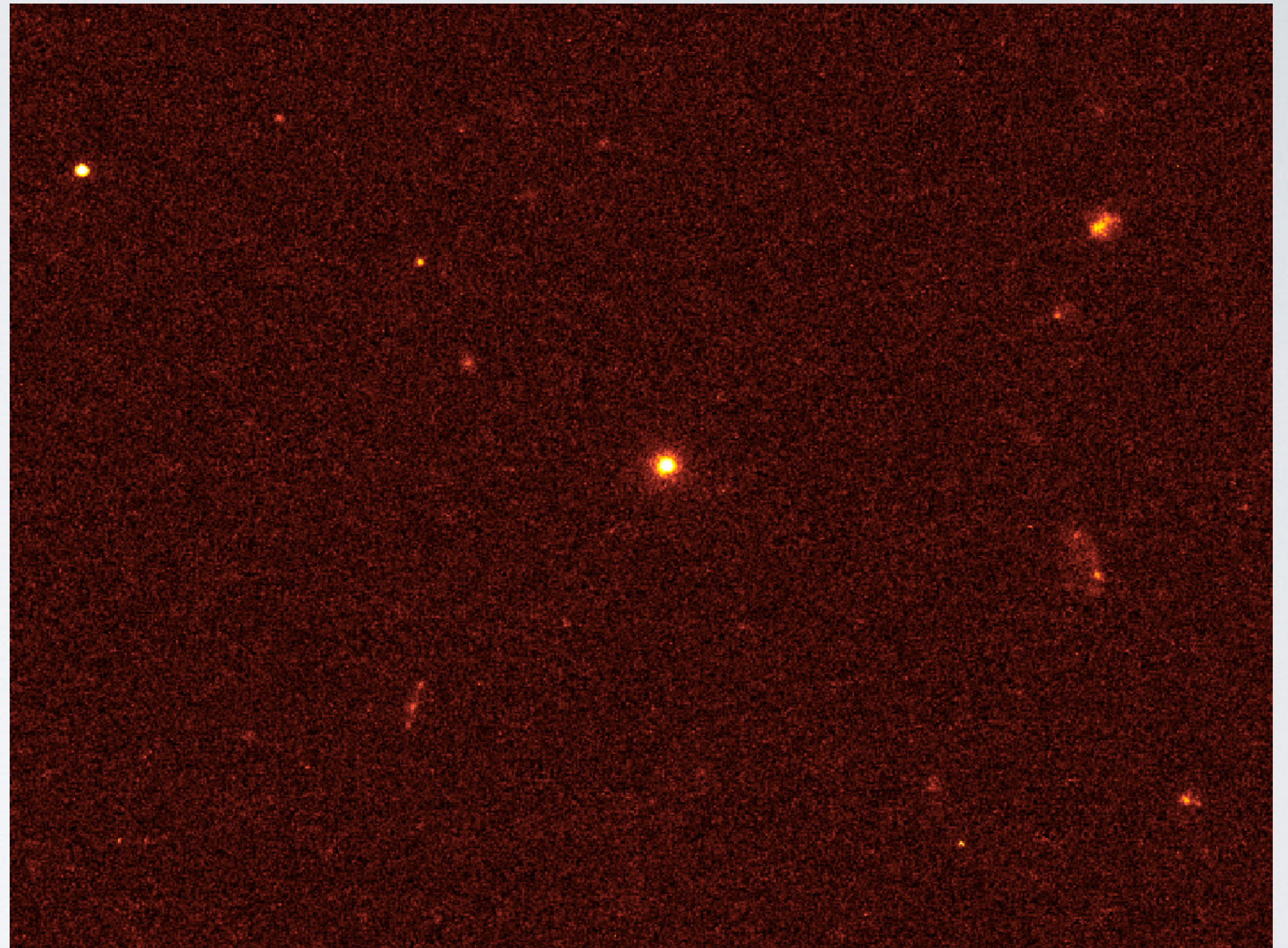


SPAGHETTIFIED STAR RIPPED APART BY A BLACK HOLE LIGHTS UP WITH A RELATIVISTIC JET

TIDAL DISRUPTION EVENTS

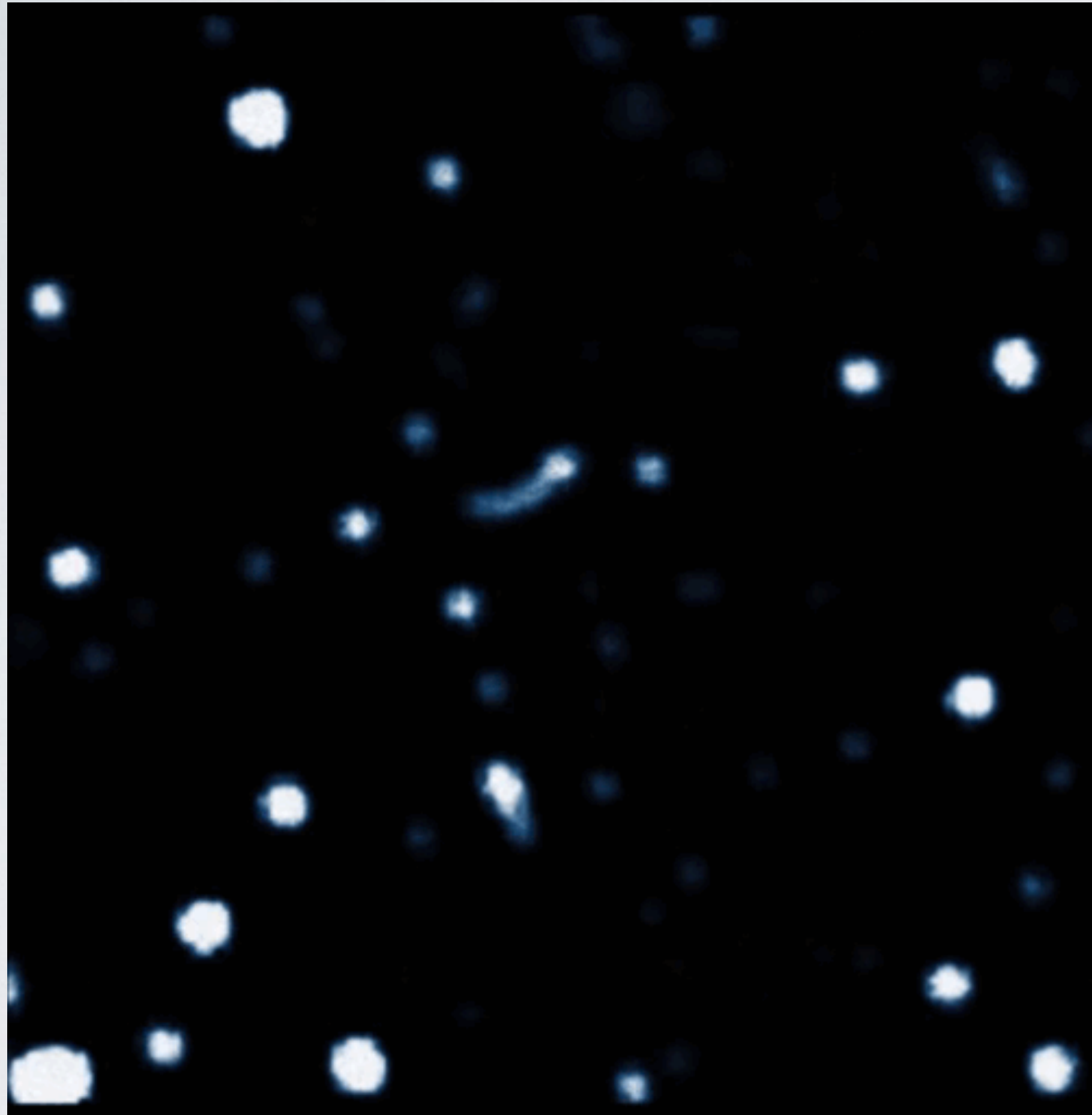


Stars circa
1.19 billion km away from
black hole
(1700 solar radii)

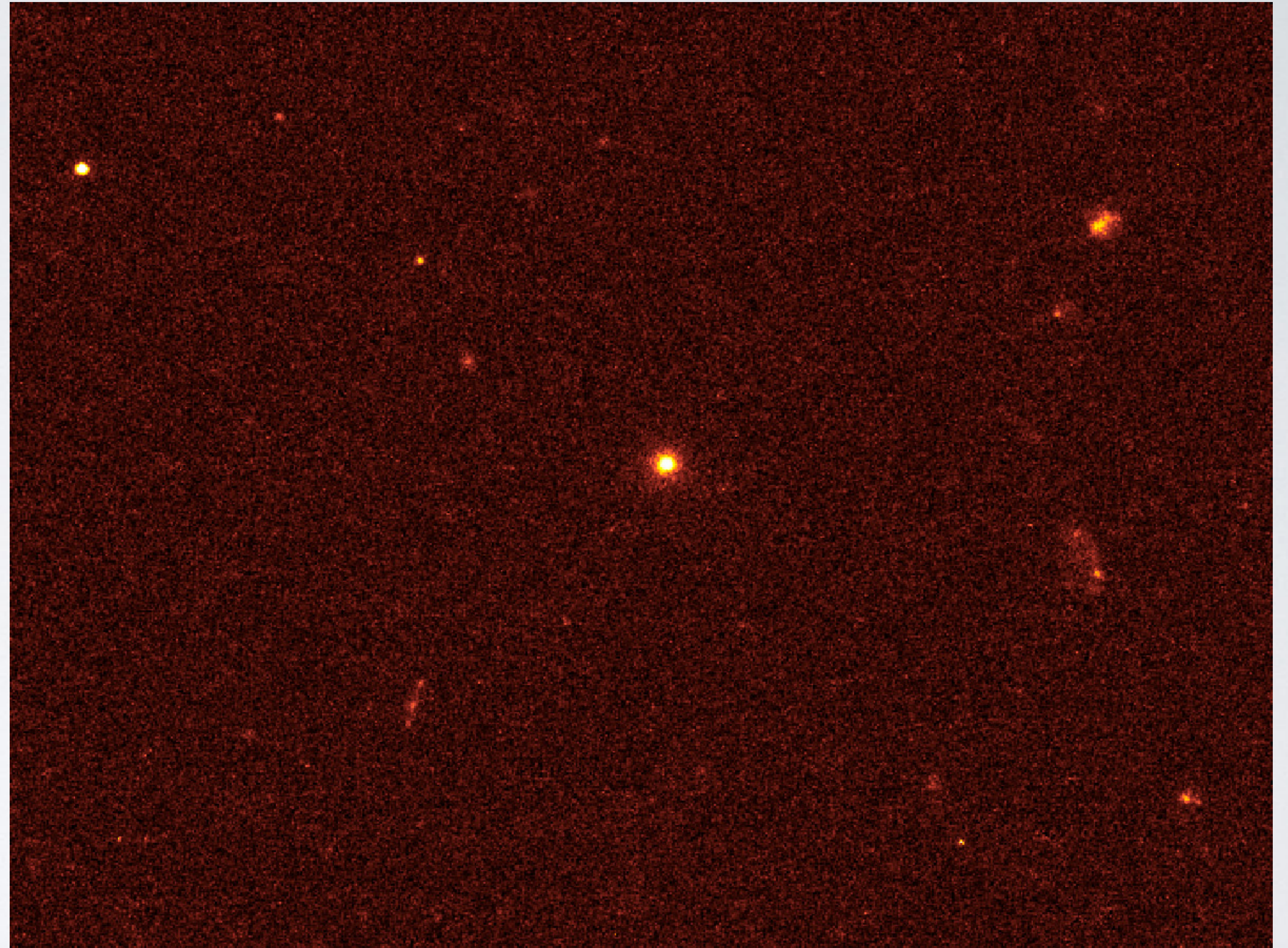


Stars circa
70 million km away from
black hole
(100 solar radii)

TIDAL DISRUPTION EVENTS

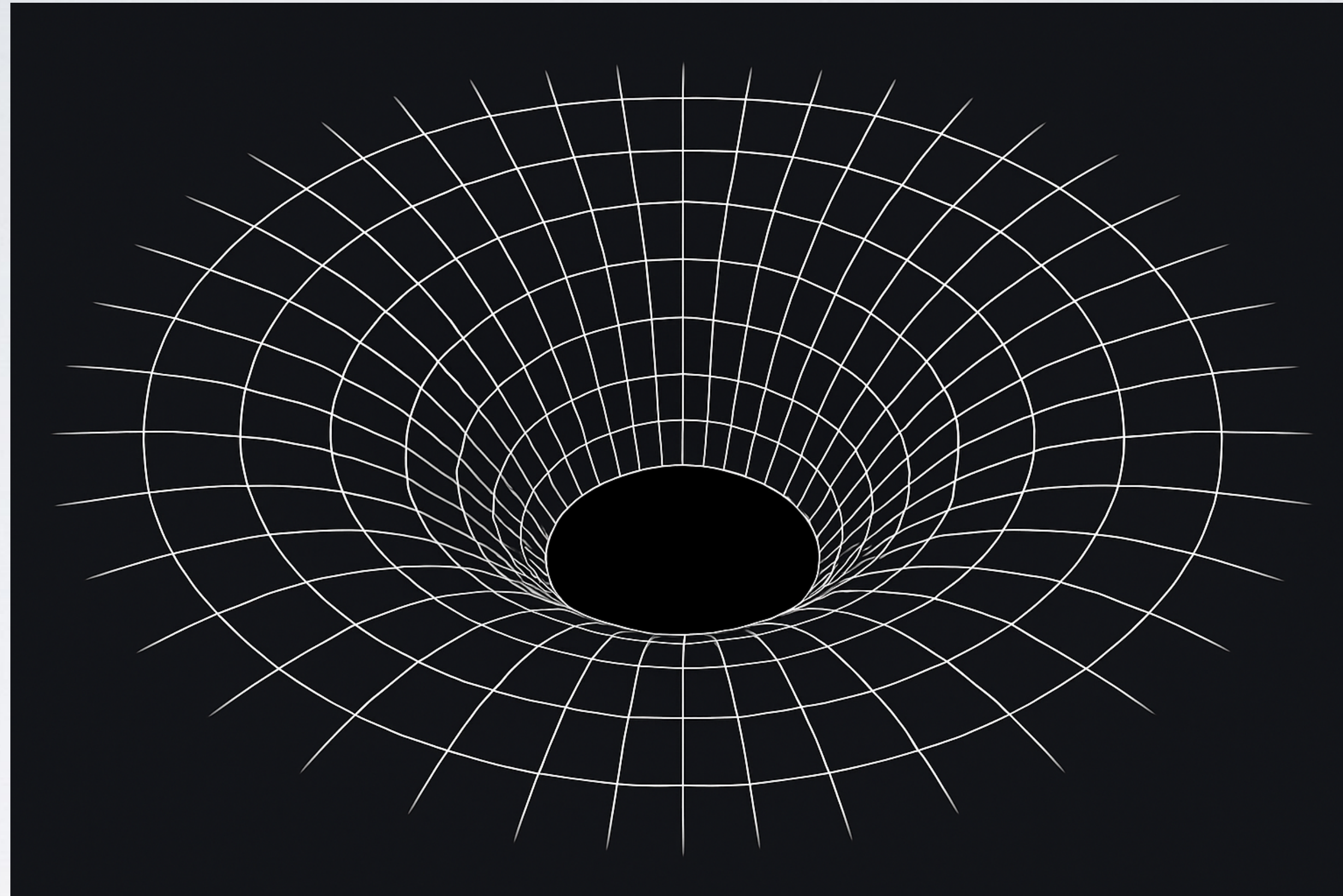


Stars circa
1.19 billion km away from
black hole
(1700 solar radii)



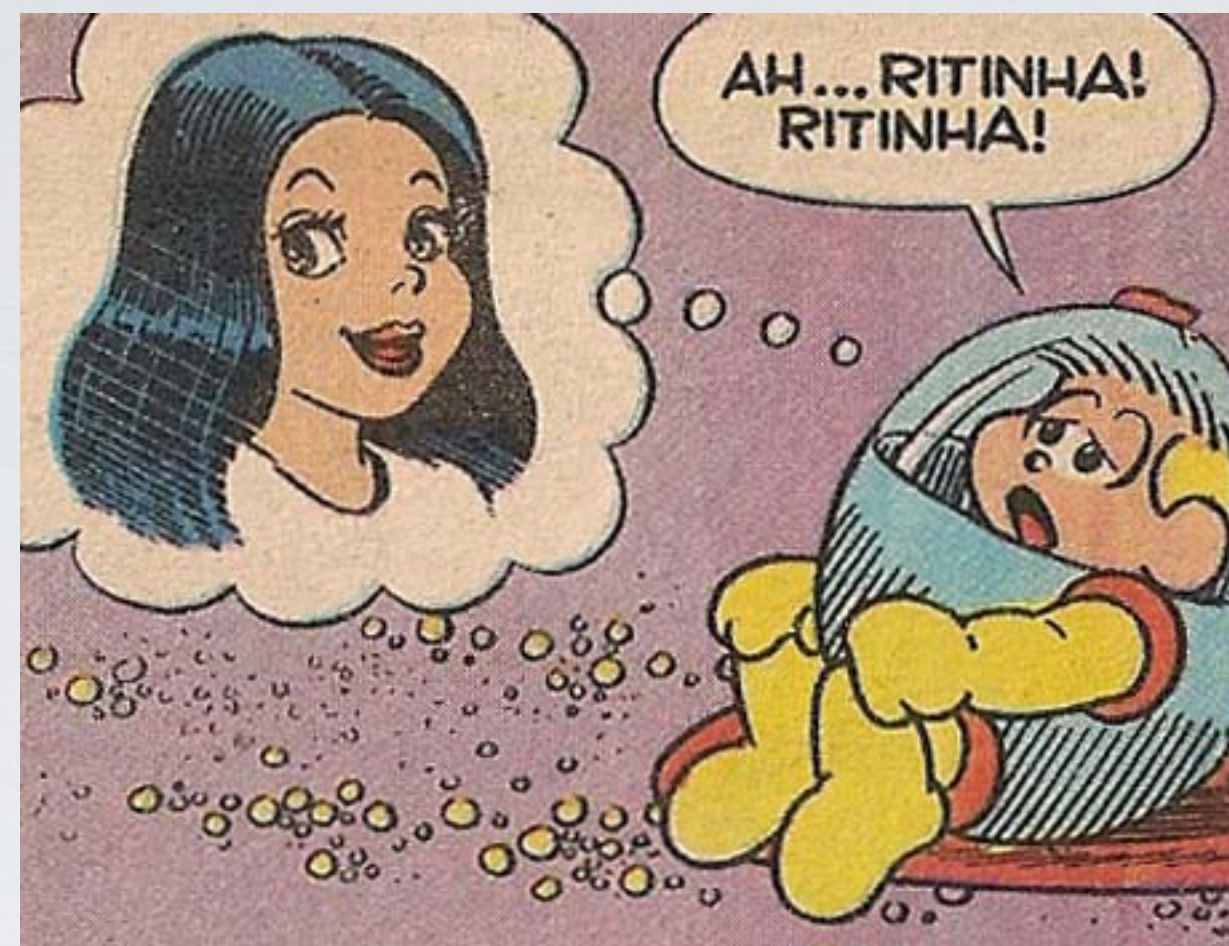
Stars circa
70 million km away from
black hole
(100 solar radii)

INSIDE THE BLACK HOLE?



“Spacetime tells matter how to move; matter tells spacetime how to curve”

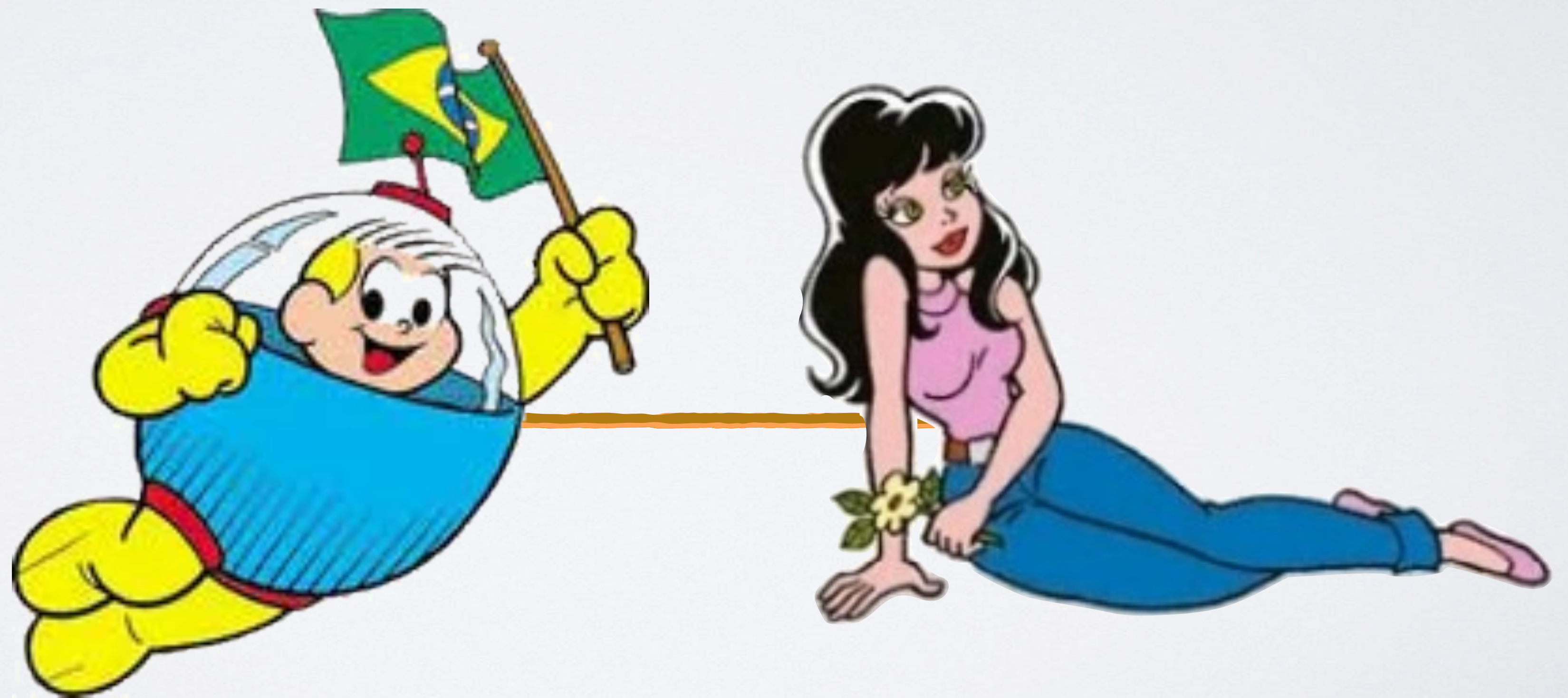
(John. A. Wheeler)



I will send you a message
everyday when I wake up
(my clock is 8am)







send **everyday**
at 8:00,00 am



receive **everyday**
at 8:00,01 am





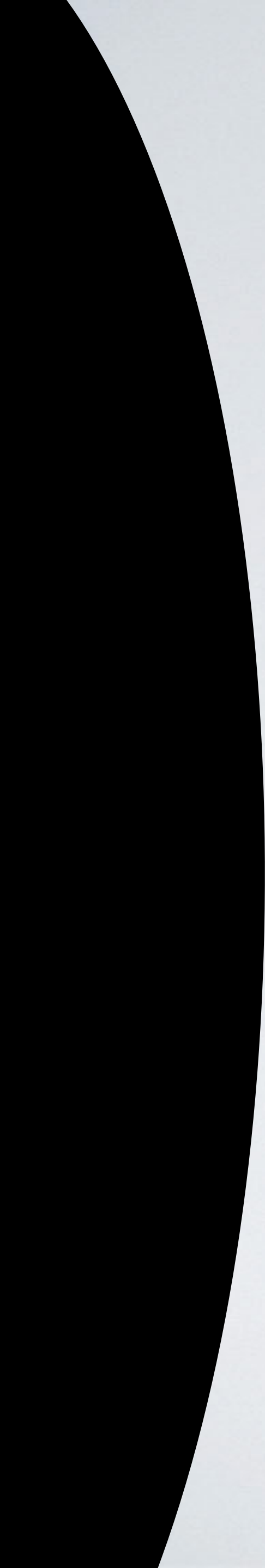


send **everyday**
at 8:00,00 am

receive **everyday**
at 02:00,0 pm





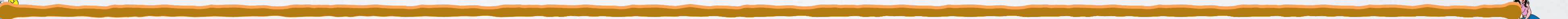
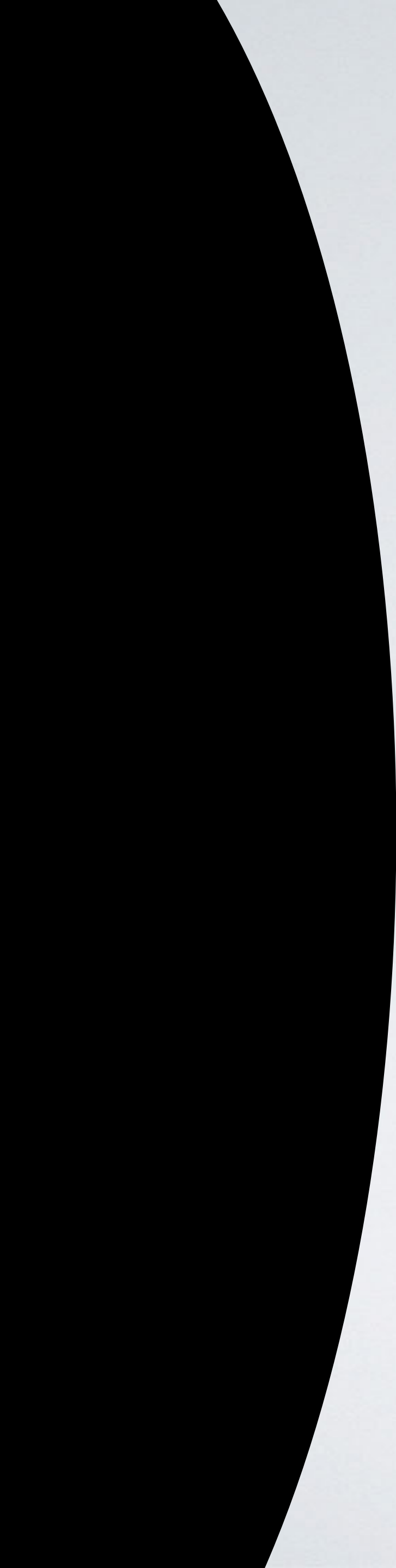


send **everyday**
at 8:00,00 am

receive **every two days**
at 05:00,0 pm







send **everyday**
at 8:00,00 am



send **everyday**
at 8:00,00 am



receive **every week**
at 11:00,0 pm







send **everyday**
at 8:00,00 am



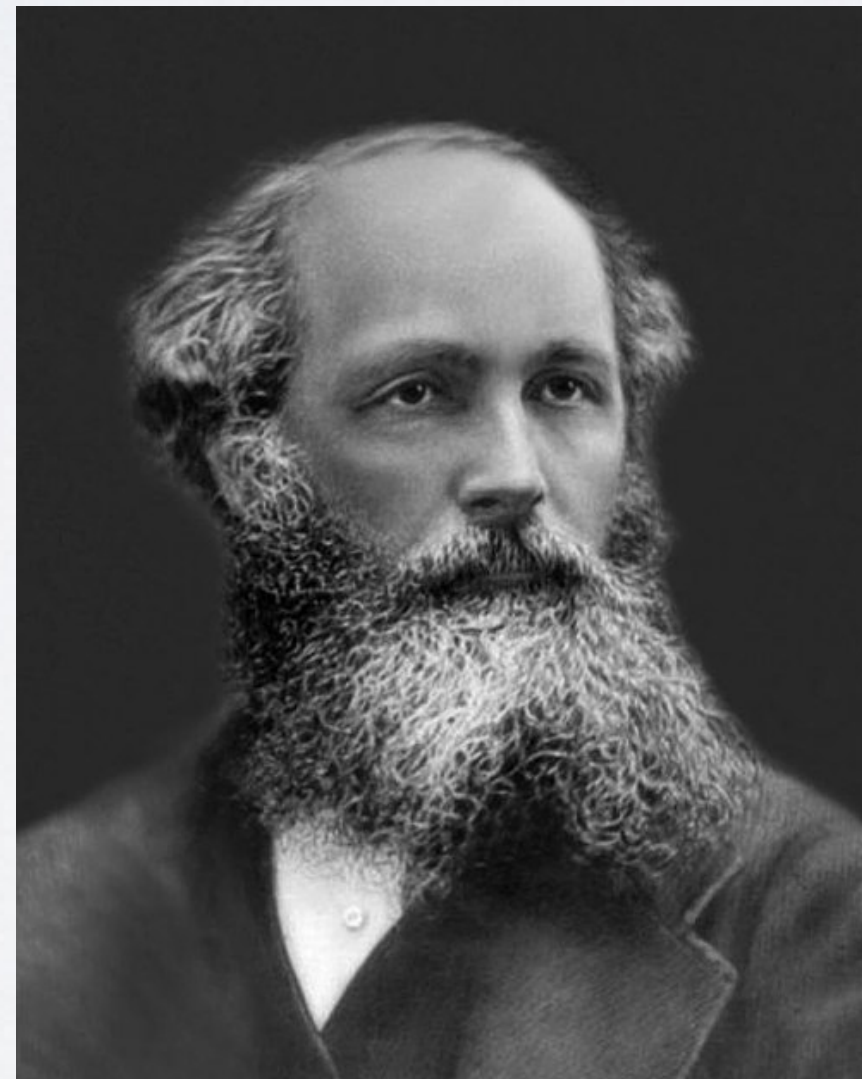
receive **never**



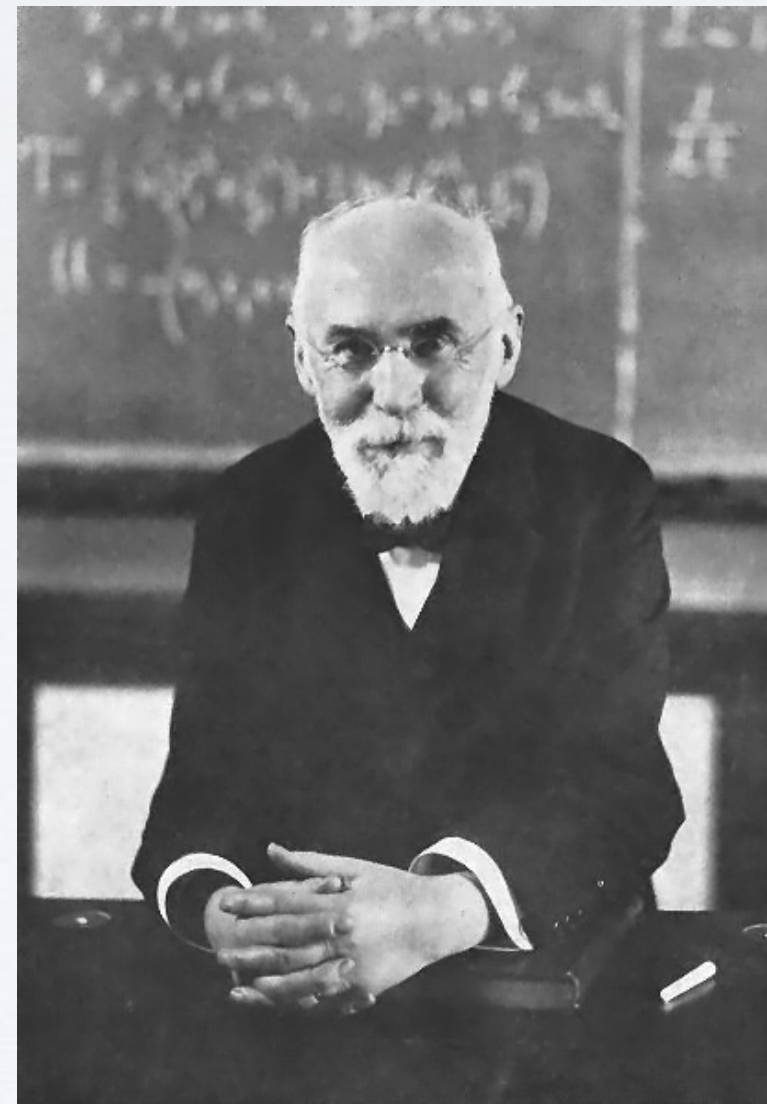
SPACE AND TIME **VERSUS** SPACETIME



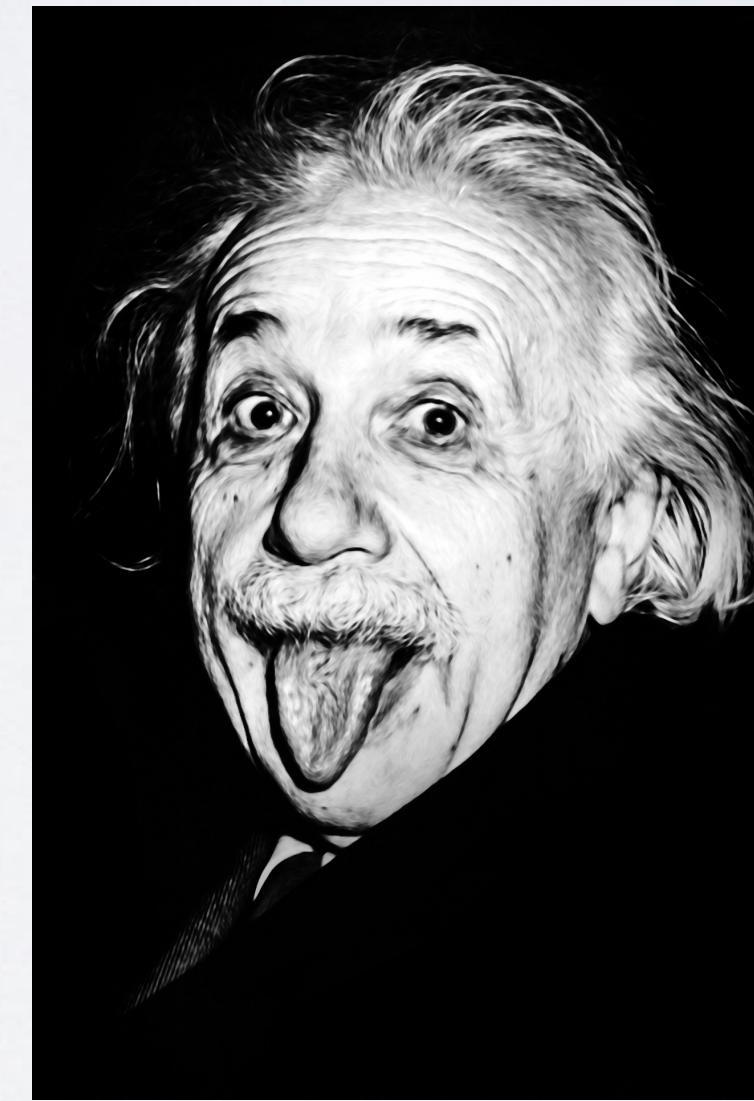
Isaac Newton (by G. Kneller)
(1643-1727)



James Clerk Maxwell
(1879-1995)



Hendrik Antoon Lorentz
(1853-1928)



Albert Einstein
(1879-1955)



Herman Minkowski
(1864-1909)

SPACE AND TIME VERSUS SPACETIME

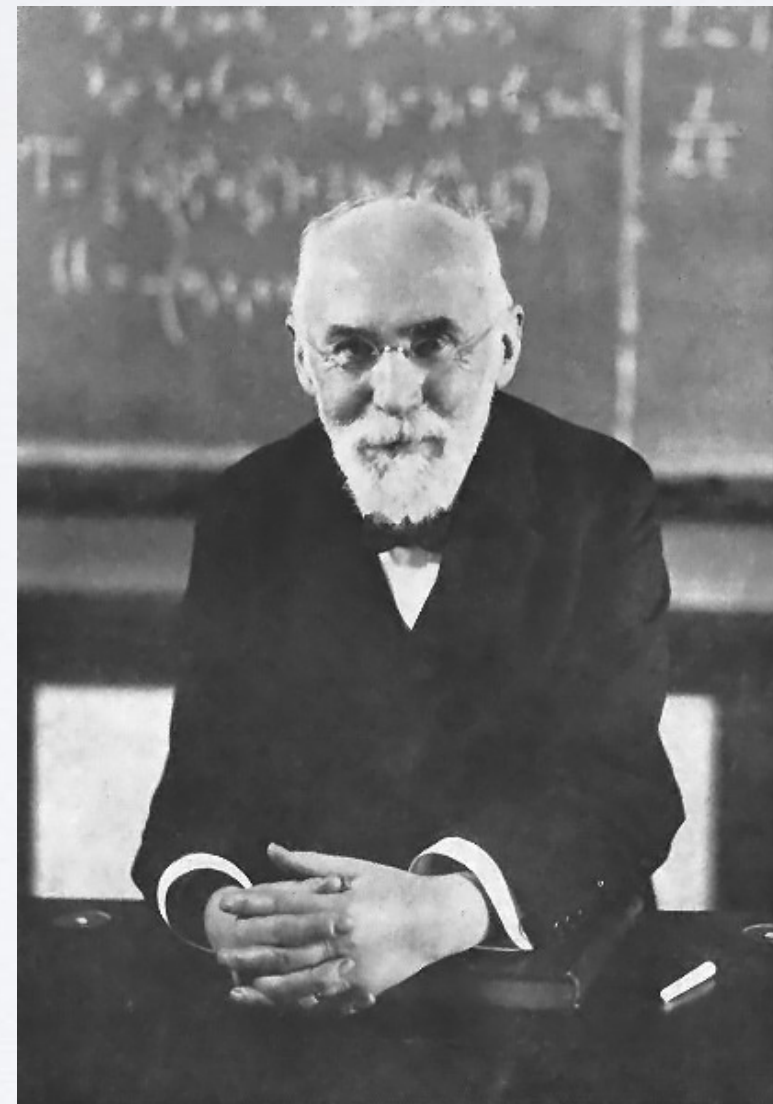
TO BE CONTINUED



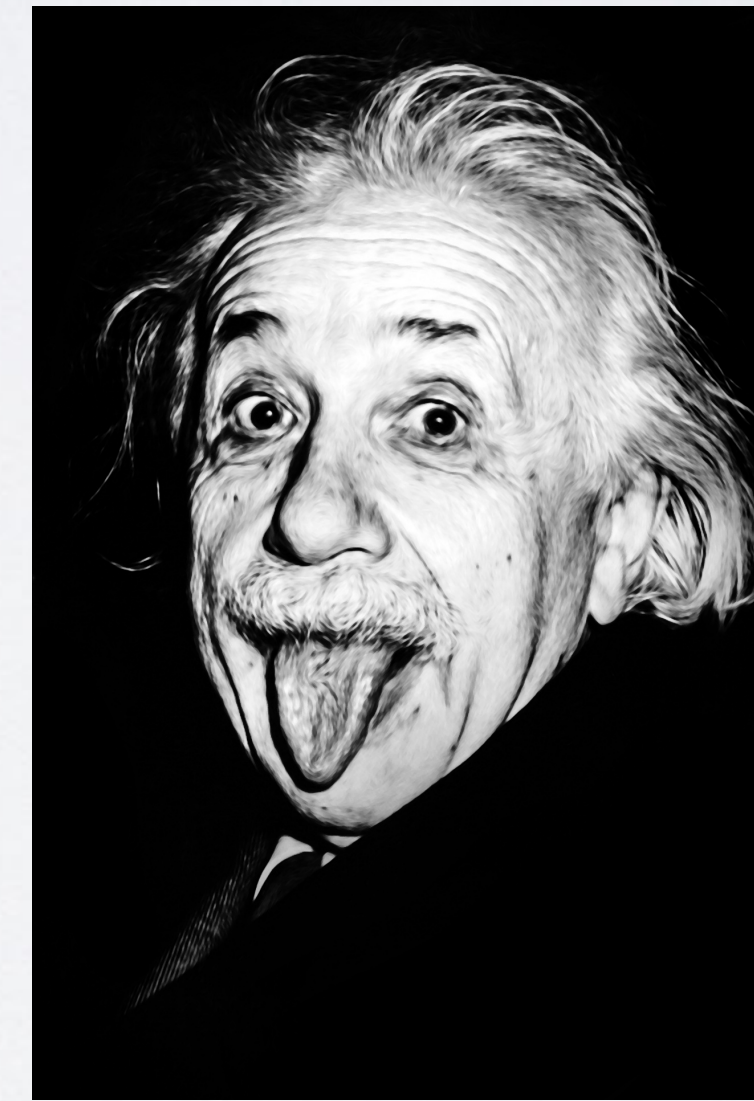
Isaac Newton (by G. Kneller)
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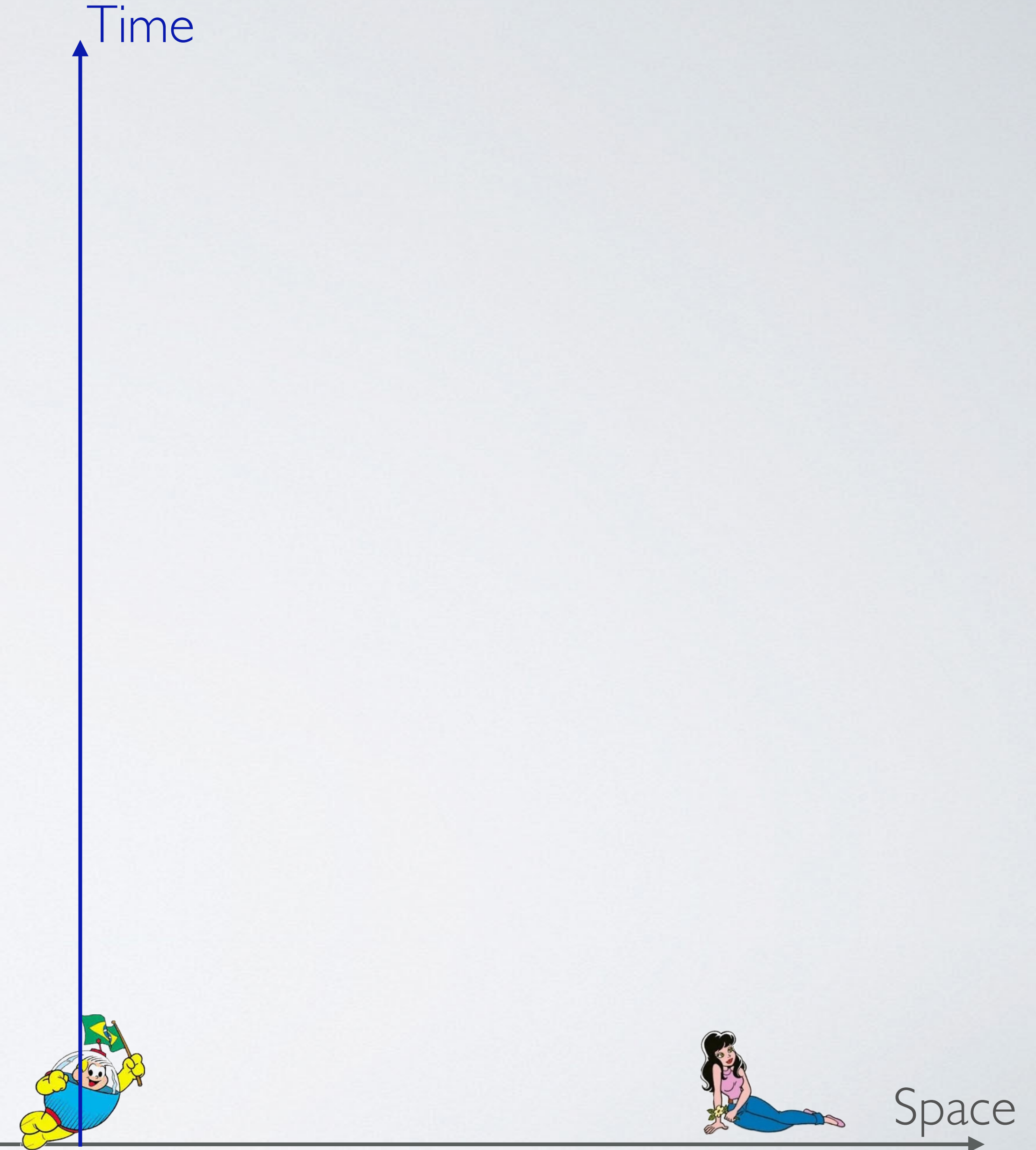
SPACETIME CAUSAL STRUCTURE



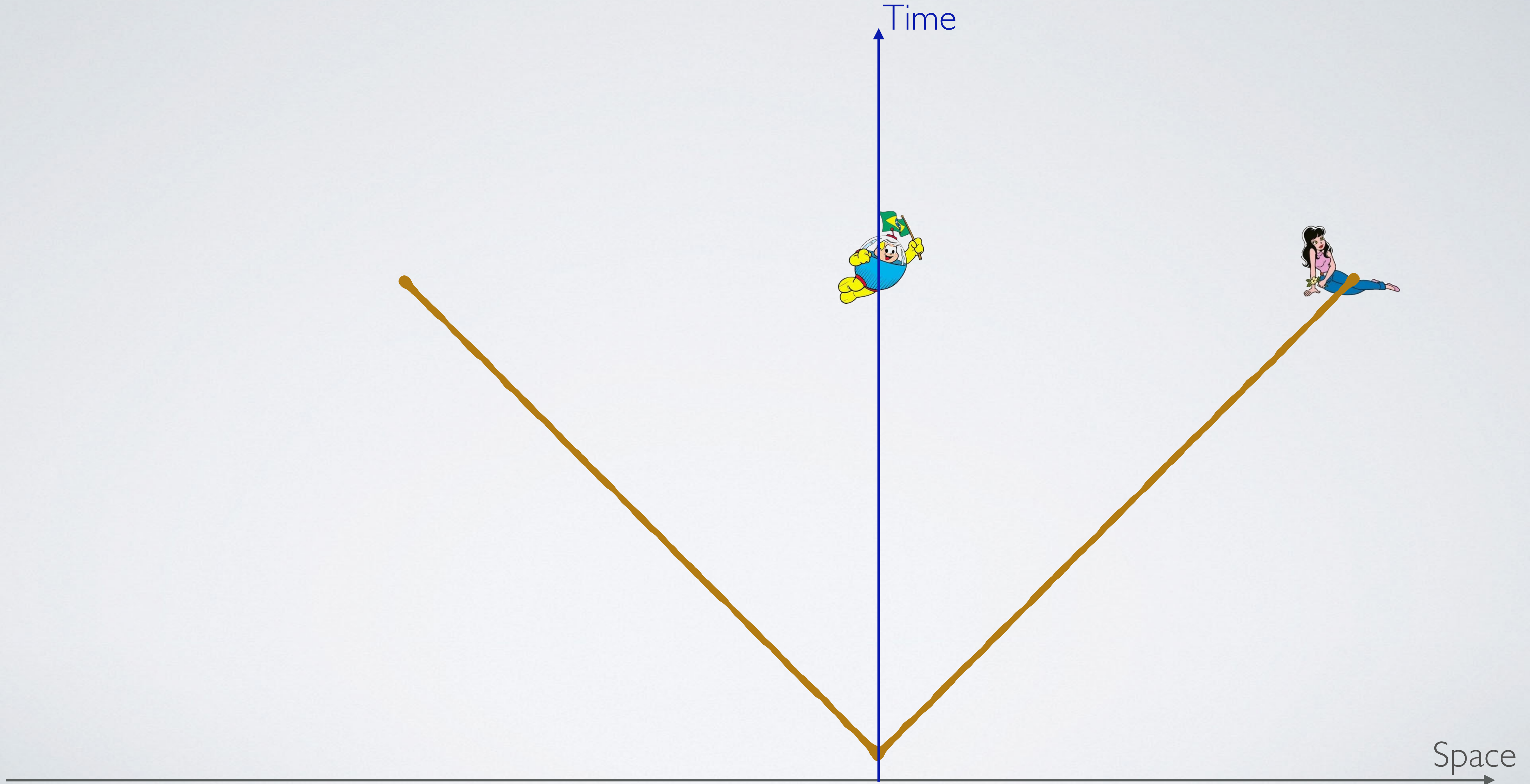
SPACETIME CAUSAL STRUCTURE



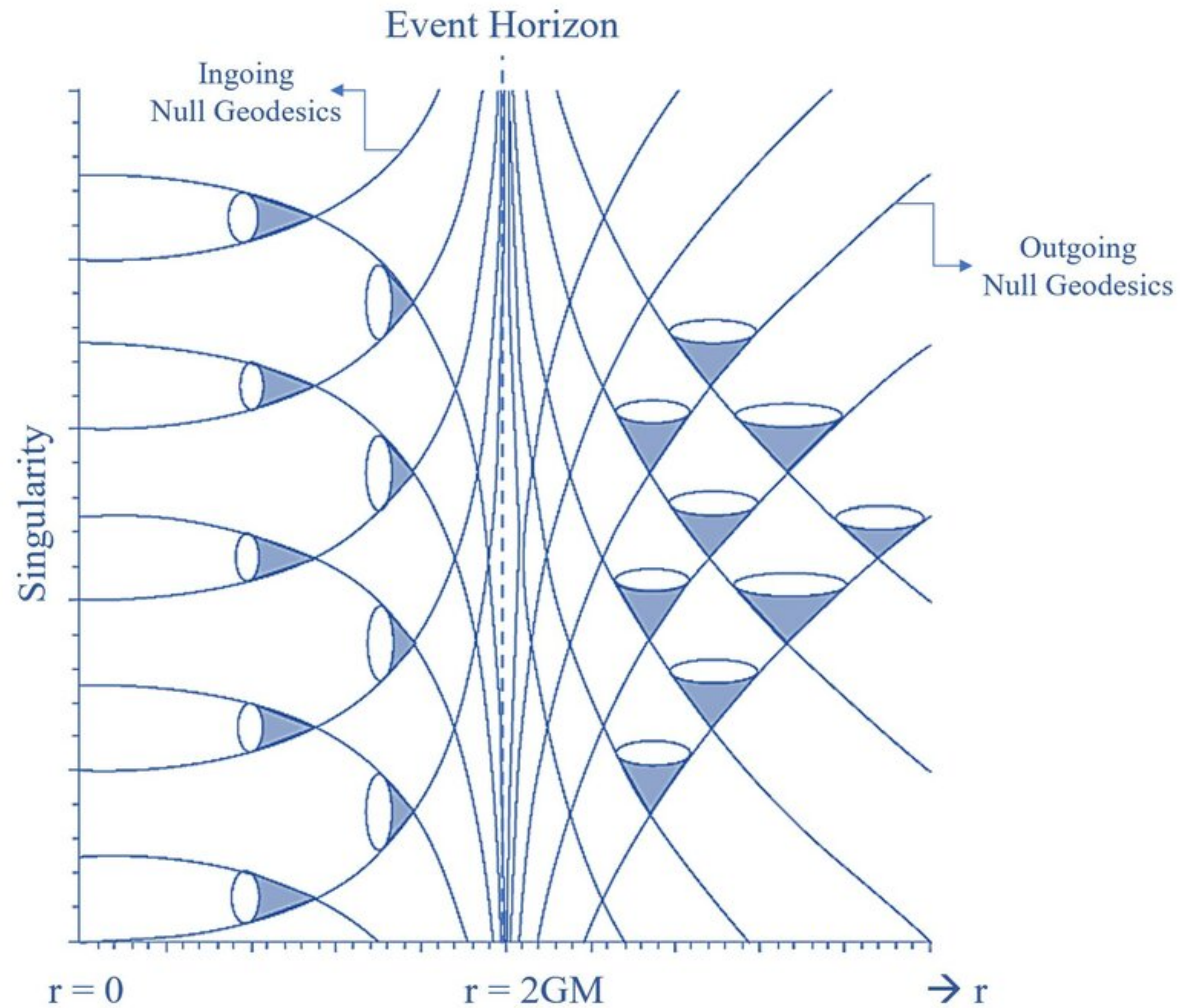
SPACETIME CAUSAL STRUCTURE



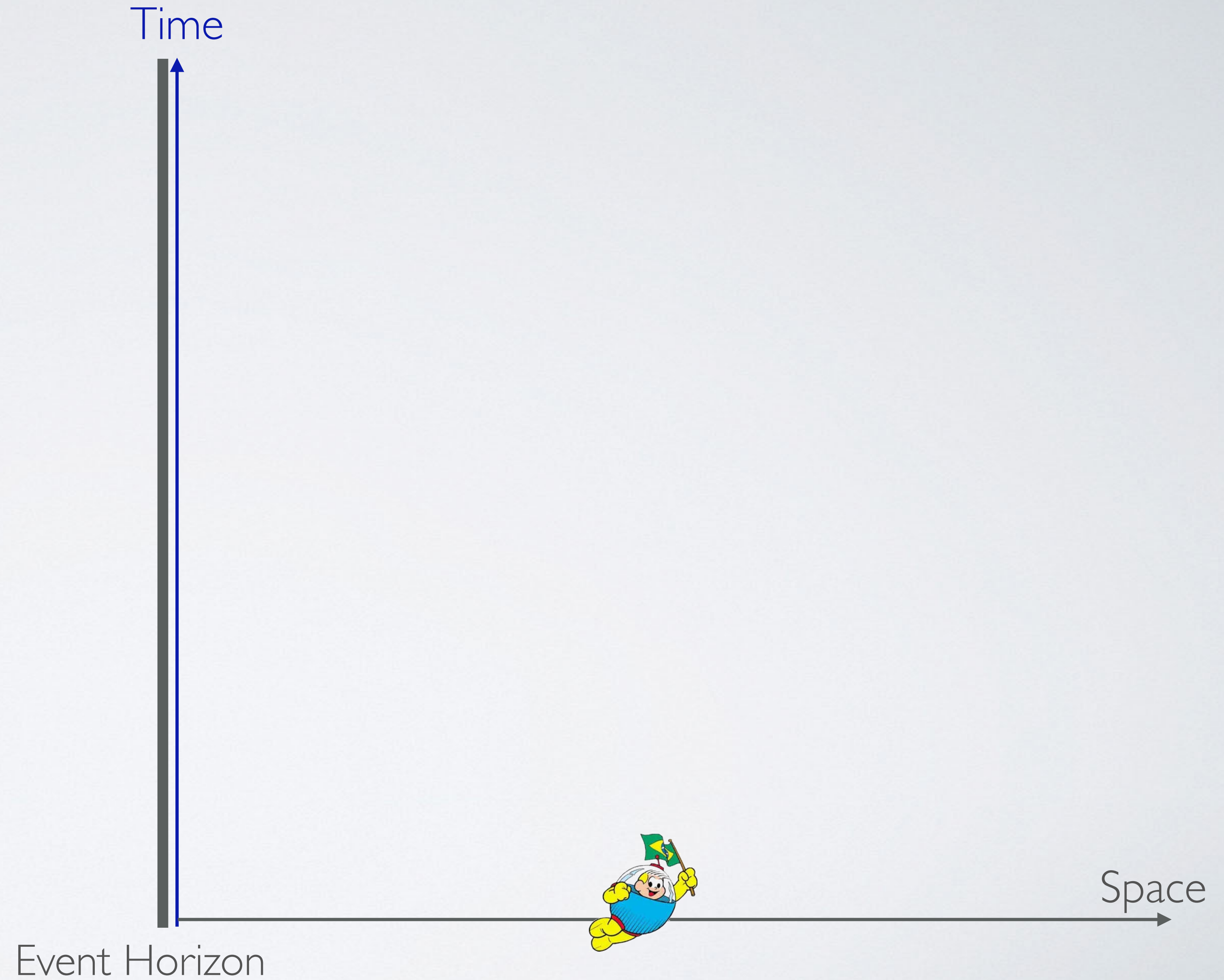
SPACETIME CAUSAL STRUCTURE



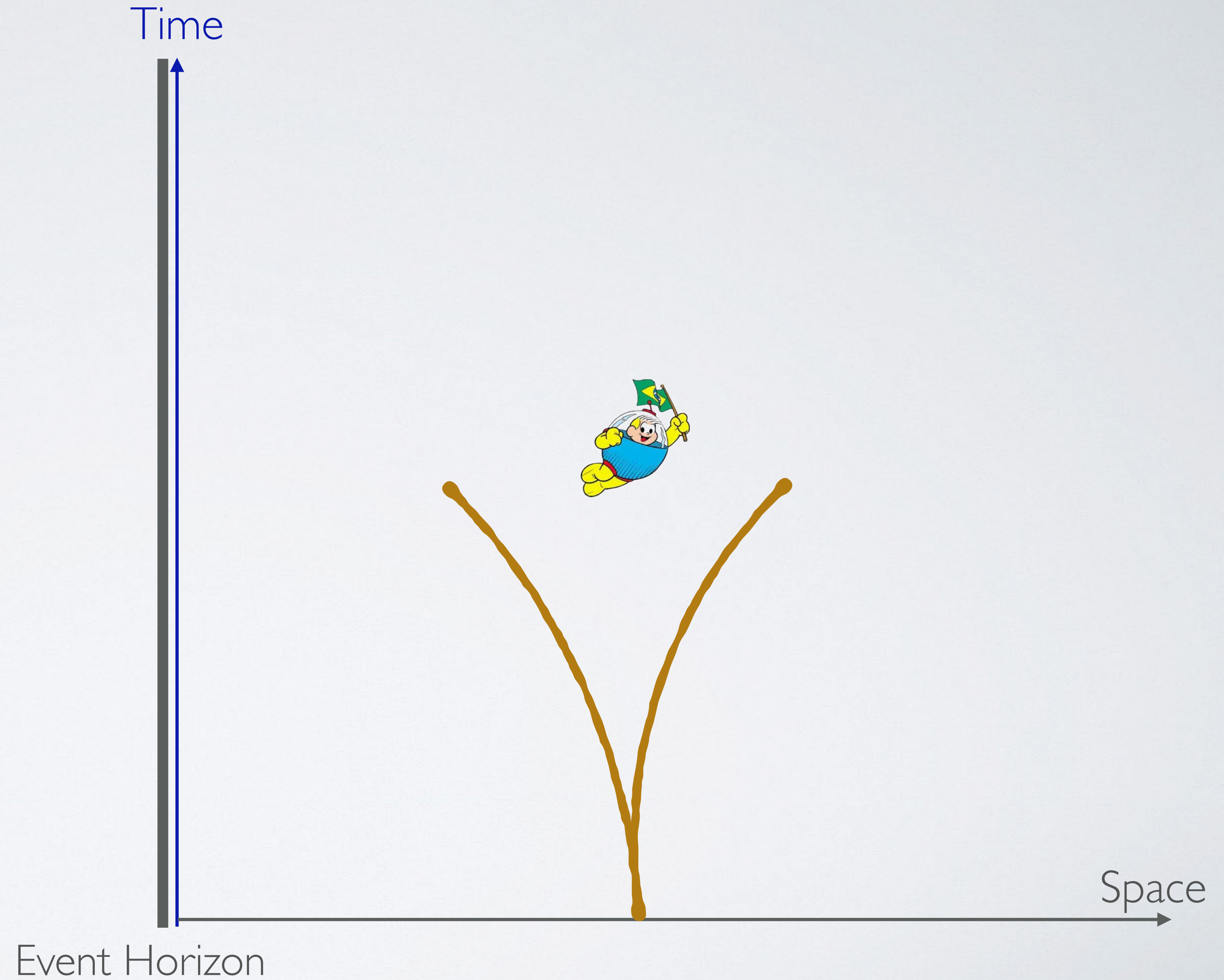
SPACETIME CAUSAL STRUCTURE



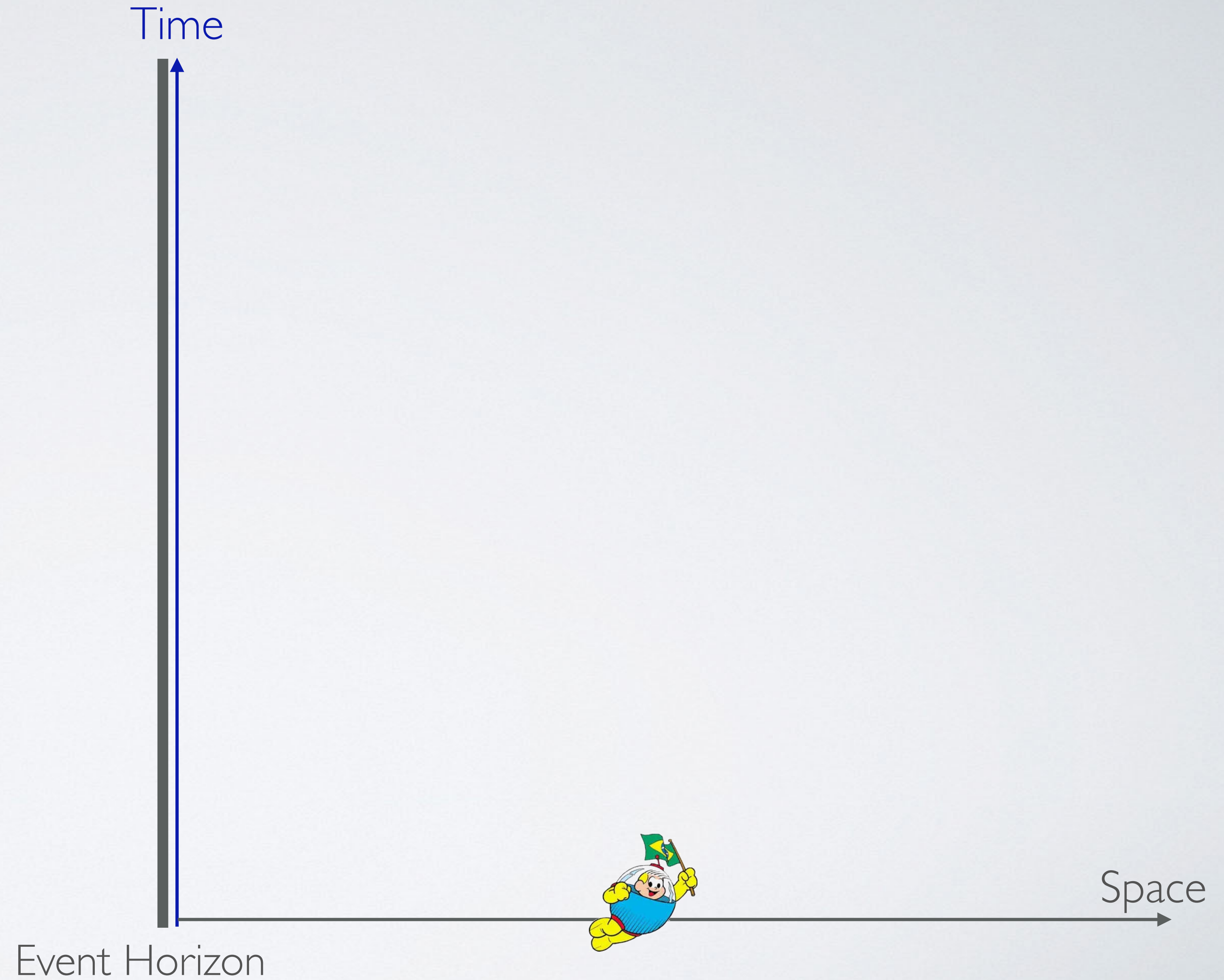
SPACETIME CAUSAL STRUCTURE



SPACETIME CAUSAL STRUCTURE



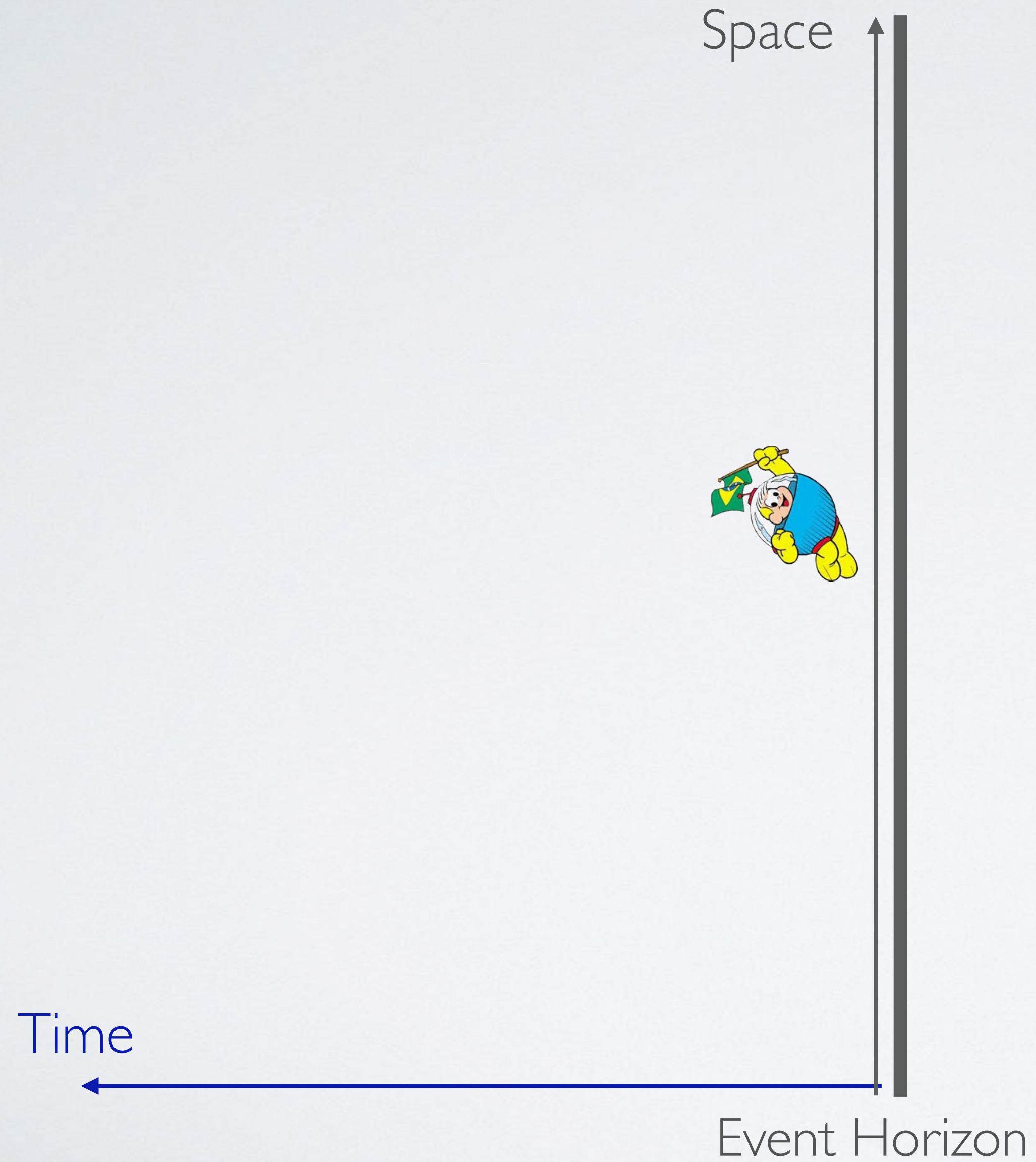
SPACETIME CAUSAL STRUCTURE



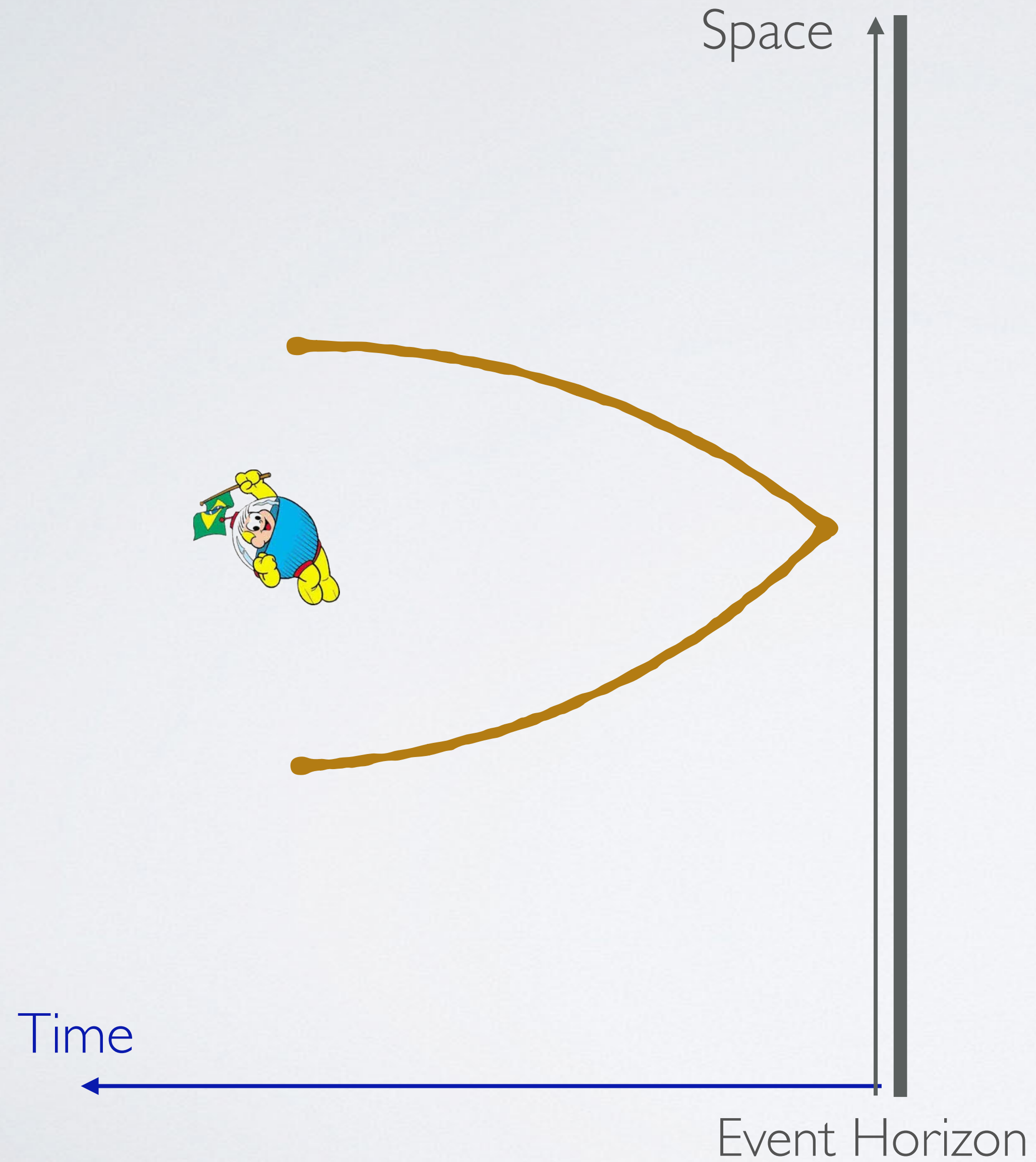
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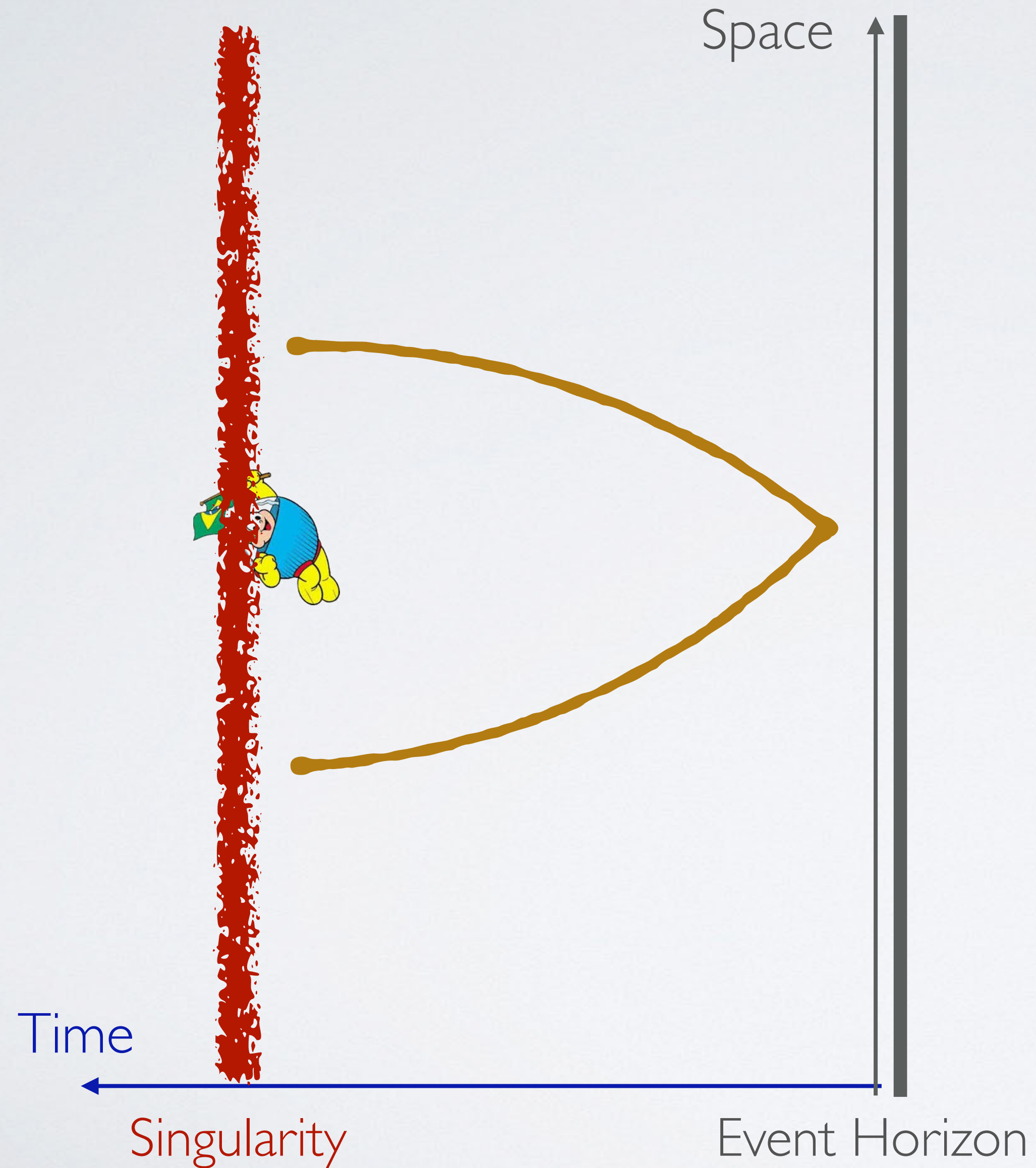
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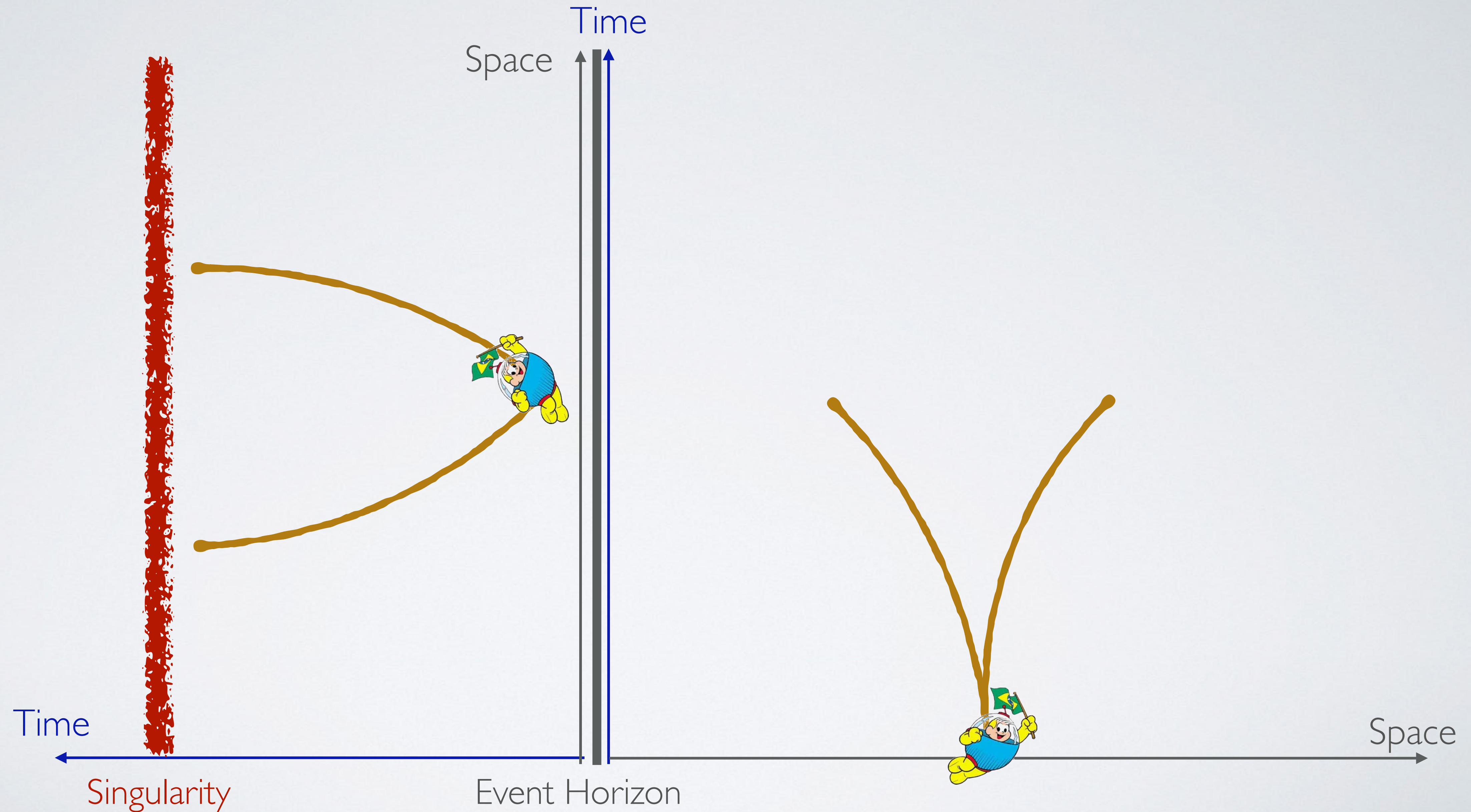
SPACETIME CAUSAL STRUCTURE



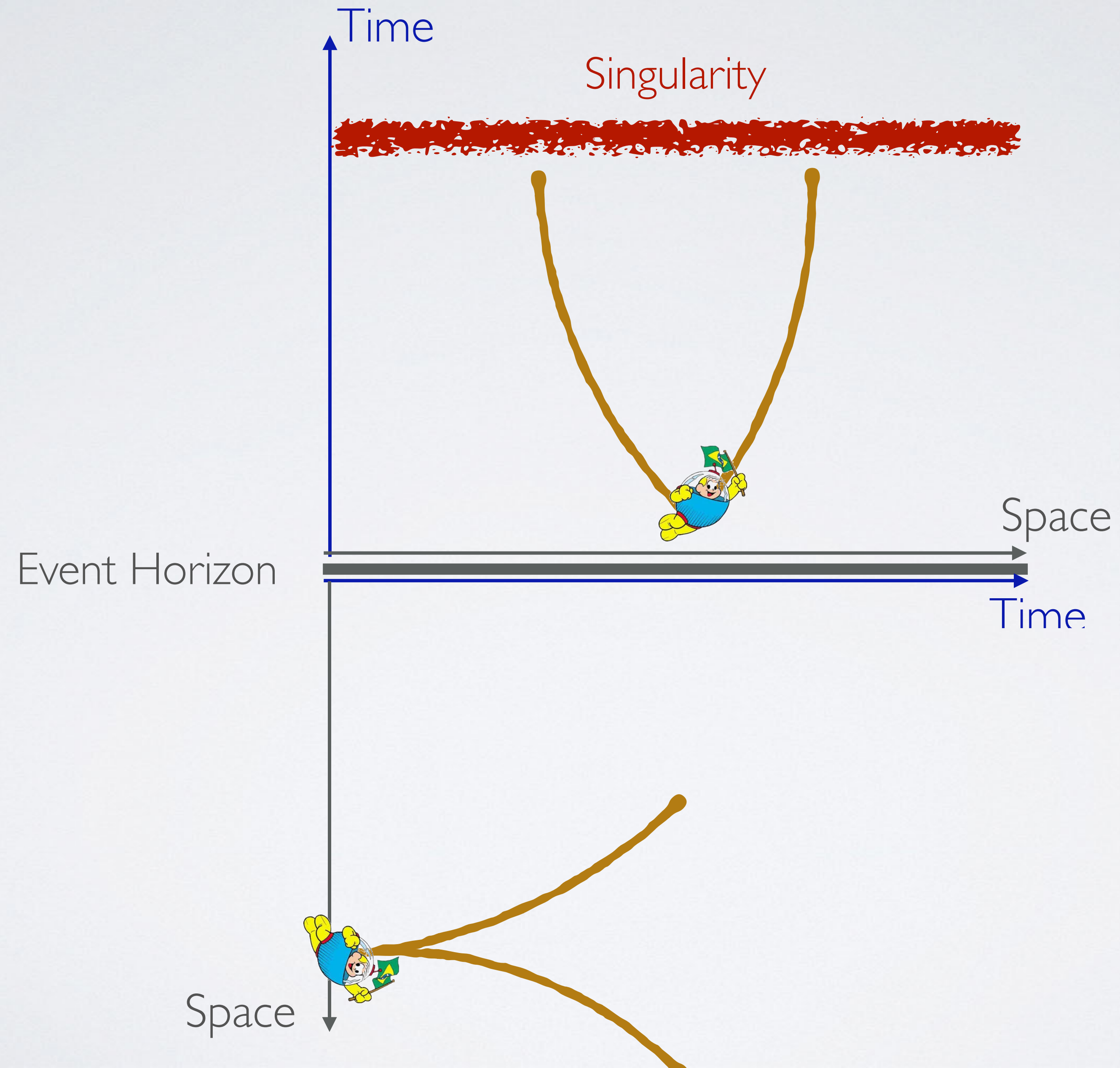
SPACETIME CAUSAL STRUCTURE



SPACETIME CAUSAL STRUCTURE

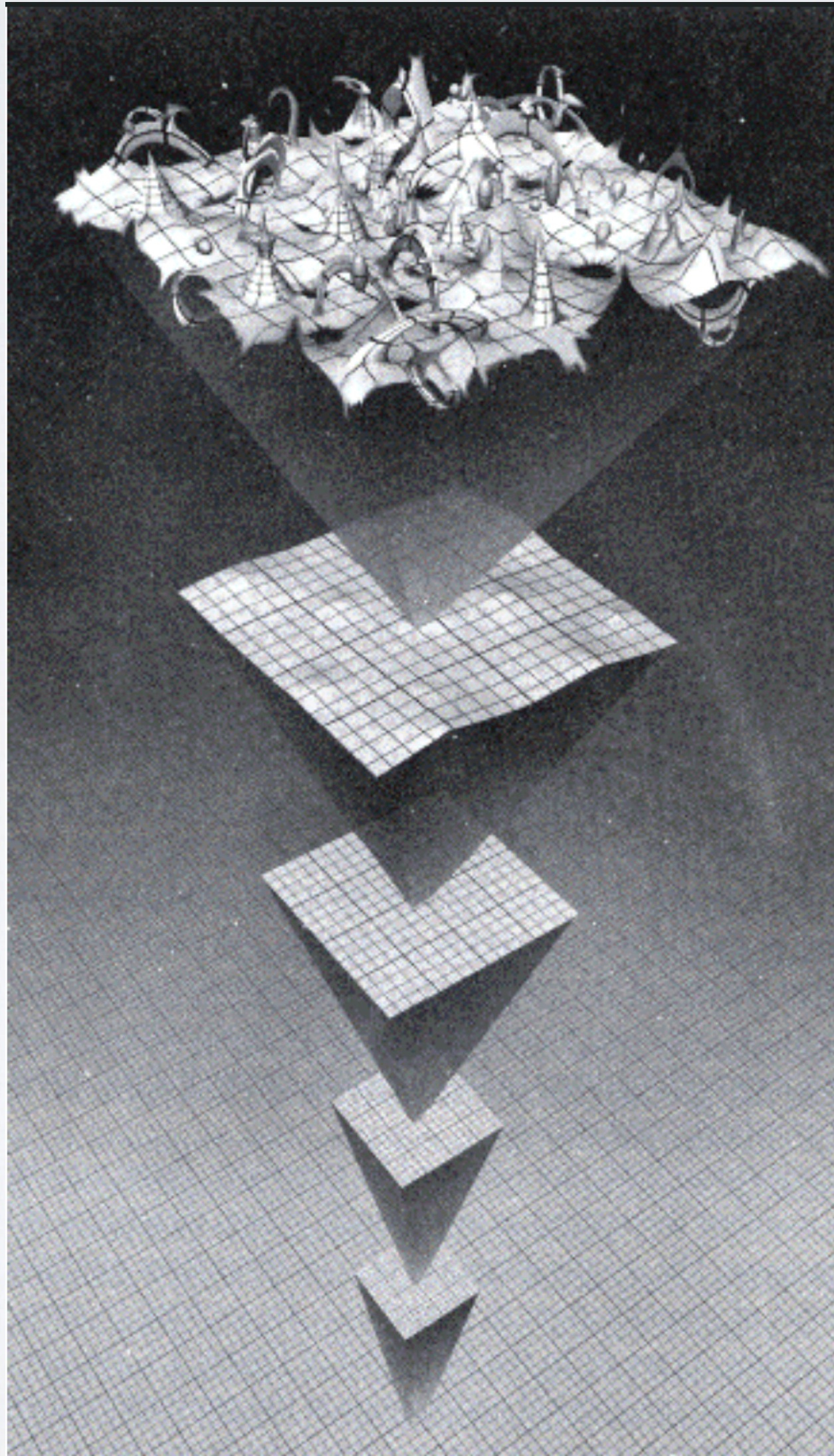


SPACETIME CAUSAL STRUCTURE



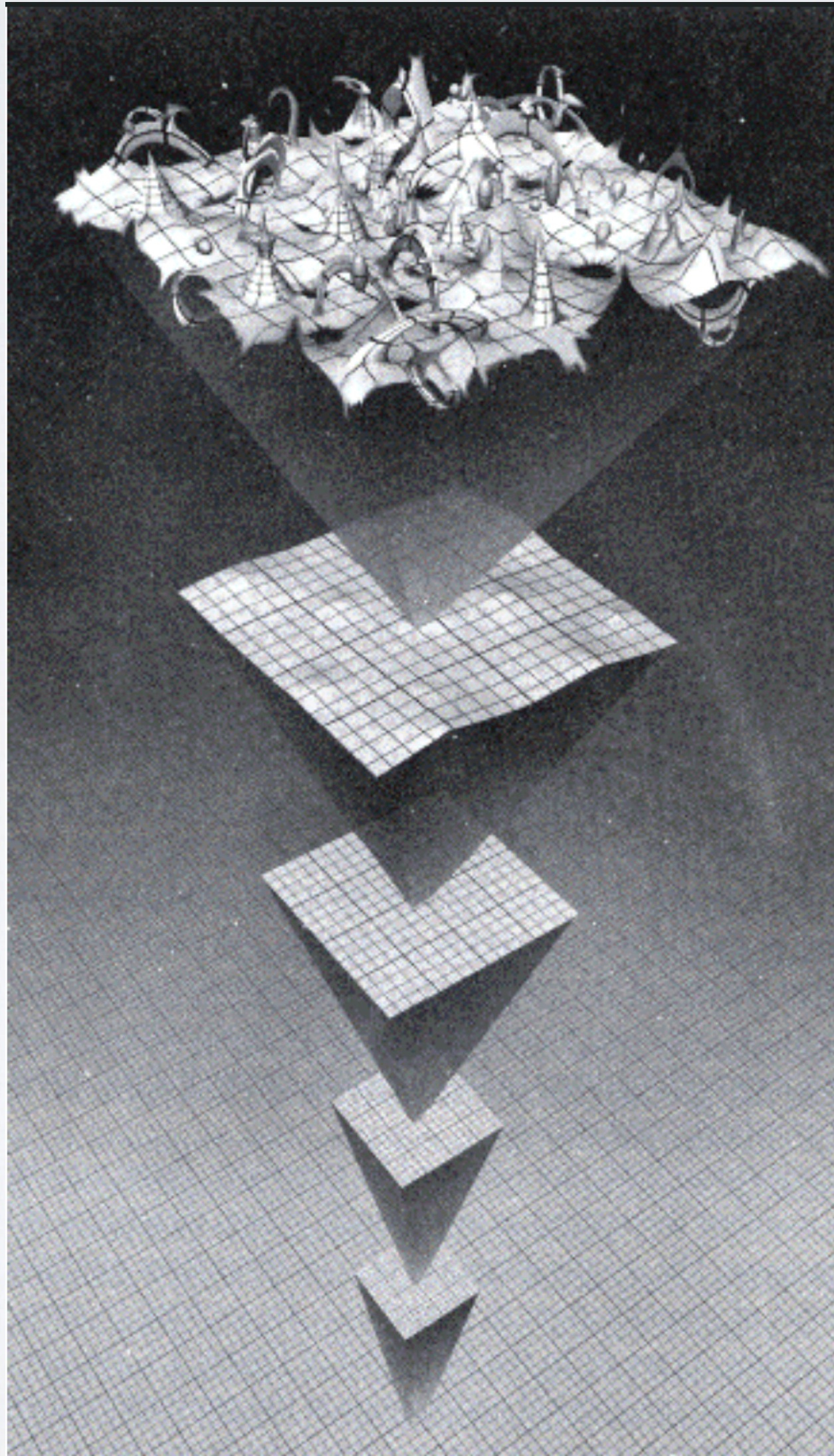
ARE BLACK HOLES REALLY BLACK?

What happens if we consider some quantum effects?



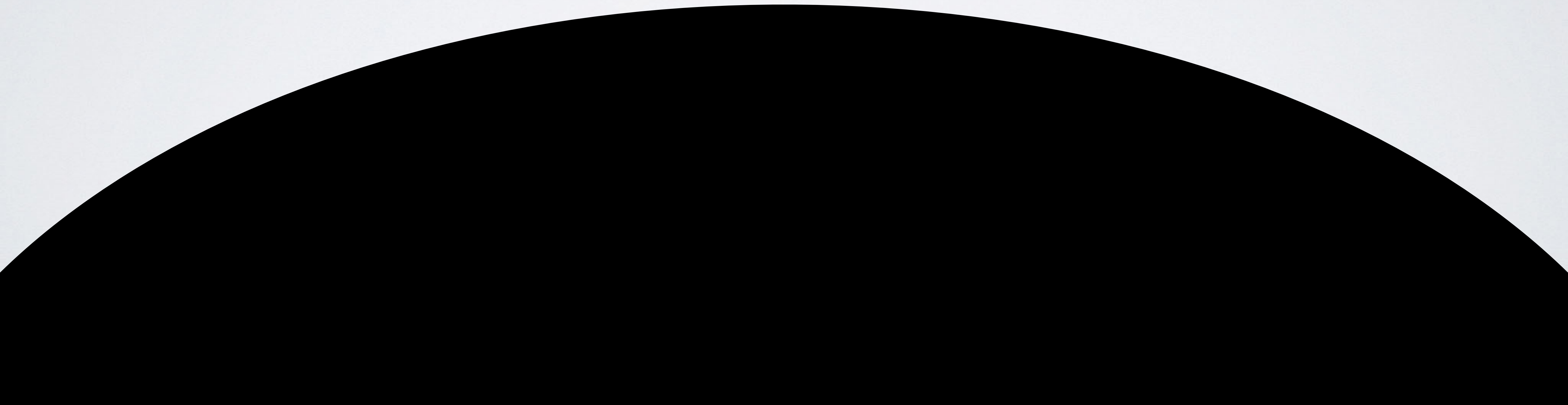
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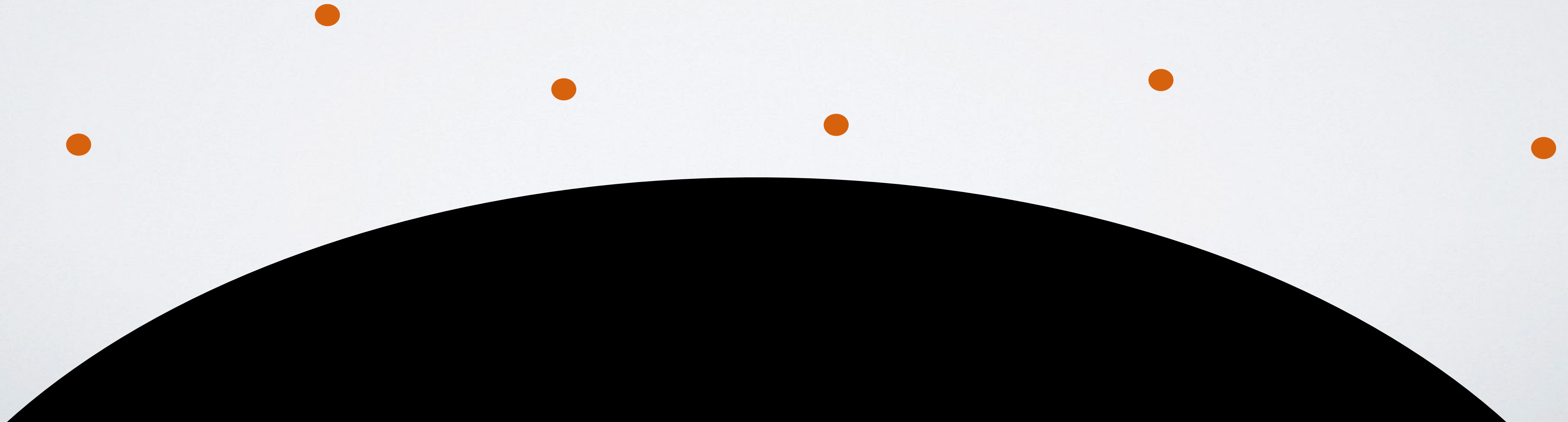
ARE BLACK HOLES REALLY BLACK?

Vacuum fluctuation at the event horizon!



ARE BLACK HOLES REALLY BLACK?

Vacuum fluctuation at the event horizon!



ARE BLACK HOLES REALLY BLACK?

Vacuum fluctuation at the event horizon!

